Fixing FOLIO and MALLS: Verified Annotations and an LLM-assisted Framework to Focus Human Relabeling

Andrea Brunello[1] **Cristian Curaba**[1] **Luca Geatti**[1]

Michele Mignani[1*] **Angelo Montanari**[1] **Nicola Saccomanno**[1]

[1]University of Udine, Italy
name.surname@uniud.it

[*]Corresponding author.

arXiv:2606.02837v2 [cs.CL] 3 Sep 2026

Abstract

Accurate translation from Natural Language to First-Order Logic (NL-to-FOL) underpins neurosymbolic AI systems and Natural Language Inference (NLI), making the quality of NL-to-FOL benchmarks essential—yet these datasets have never been rigorously audited. Our first contribution is to present a systematic human inspection of the validation split of FOLIO and a subset of MALLS test instances, finding that approximately 42.5% and 42% of entries, respectively, contain incorrect FOL formalizations (i.e., ground truth labels), with additional rates of ambiguous NL sentences (17.8% and 51%) and incorrect NLI labels in FOLIO (8.4%). Our second contribution is to develop and release corrected ground truths for such datasets, showing that annotation errors distort model evaluation on a reference benchmark task: testing three state-of-the-art LLMs (Gemma 4 31B-it, Qwen3-30B-A3B, and GPT-4o-mini) with the corrected ground truths yields accuracy gains from +11 to +23 percentage points. Motivated by these findings, we propose an LLM-based framework to support humans in manual reviewing NL-to-FOL datasets. By directing reviewers toward the most error-prone instances, we empirically show that it is possible to achieve 90% dataset accuracy after reviewing fewer than 20% of instances, compared to over 76% required by unguided review. We release all human-verified annotations and the code for our framework.

1 Introduction

Automatically translating Natural Language (NL) into a machine-readable formalism—often called *autoformalization*—is a fundamental building block of neurosymbolic AI, with applications ranging from Natural Language Inference (NLI) (Ye et al., 2023; Olausson et al., 2023; Pan et al., 2023) to runtime verification and AI safety (Dalrymple et al., 2024). Among the target formalisms, First-Order Logic (FOL) stands out for its expressiveness and computational tractability. Yet, NL-to-FOL translation[1] remains a longstanding challenge for both humans and automated systems (Barker-Plummer et al., 2009; Singh et al., 2020). We refer the reader to Brunello et al. (2026a) for a comprehensive survey of approaches, datasets, and open challenges in NL-to-FOL formalization, and we will adopt the terminology introduced therein throughout this paper.

To support progress in this area, datasets have been developed to facilitate training of NL-to-FOL translation systems and enable their systematic comparison. In practice, however, only a limited number of partially human-curated resources are publicly available, most notably FOLIO (Han et al., 2024) and MALLS (Yang et al., 2024a). Prior work has examined limitations in evaluation metrics and task design (Brunello et al., 2026b), but the reference annotations themselves have received little systematic attention.

In this paper, we show that these annotations contain significant errors. We conduct a systematic human audit of the FOLIO validation split and a subset of MALLS test instances, finding that 42.5% and 42% of examples, respectively, contain incorrect FOL formalizations. Additionally, 17.8% of FOLIO and 51% of MALLS include NL statements that admit multiple, non-equivalent yet defensible interpretations (Section 2). These annotation errors have a direct and notable impact on evaluation: re-assessing state-of-the-art models (Gemma 4 31B-it, Qwen3-30B-A3B, GPT-4o-mini) against our corrected ground truth yields accuracy gains of +11 to +23 percentage points (Section 2.3).

Exhaustive manual review, however, does not scale to large datasets or real-world deployments. We therefore introduce an LLM-assisted oversight framework that focuses human effort on the in-

[1]See Appendix A for a more formal definition.

stances most likely to be incorrect. The approach exploits a key empirical finding: LLMs rarely flag a correct formalization as wrong—in the FOLIO validation split, Gemma marks only $\sim$3% of correct instances as incorrect (see Appendix I). Directing annotators to instances that the model deems incorrect, our framework enables 90% annotation accuracy while requiring review of only $\sim$20% of FOLIO validation instances and $\sim$5% of the MALLS test subset. Moreover, when applied to the high-quality private dataset GGC (Barker-Plummer et al., 2011b), it introduces negligible additional noise, confirming its reliability when errors are rare.

Our framework finds natural application across several settings where high-quality FOL corpora are required but exhaustive human verification is expensive. First, in *dataset curation*, rather than subjecting every instance to costly expert review, curators can focus exclusively on the small fraction flagged as suspicious, turning the process into a much more tractable task. Second, in *formal methods* and verification, the framework can assist experts by identifying likely incorrect formalizations of NL specifications, acting as a warning layer. Finally, in *runtime verification* scenarios (Wang et al., 2025), where newly defined behavioral constraints must be checked immediately against live data, there is no time for manual review; the framework provides on-the-fly feedback on whether a generated formula faithfully captures the intended requirement.

Contributions.

1. **Corrected datasets:** revised FOL formalizations and NLI labels (when applicable) for the FOLIO validation split (275 instances) and for a MALLS test subset (100 instances) (Section 2).

2. **Evaluation impact:** accuracy gains (from +11 to +23 percentage points) when state-of-the-art LLMs are assessed against the corrected ground truth (Section 2.3).

3. **LLM-assisted oversight framework:** a novel strategy that directs human annotation effort toward instances most likely to contain errors, substantially reducing review cost without sacrificing quality of the revision (Section 3).

We release the corrected datasets and all code of our oversight framework at `https://github.com/dslab-uniud/Fixing-FOLIO-and-MALLS.git`.

2 Datasets and Their Quality Analysis

We analyze errors and ambiguities in two datasets splits routinely used by the community for NL-to-FOL evaluation: FOLIO validation split (Han et al., 2024) and a subset of the MALLS test set (Yang et al., 2024a). We describe in the following our methodology for human verification and correction. As shown in Table 1, approximately 42.5% of FOLIO and 42% of MALLS instances contain annotation errors, with direct consequences for model evaluation (Section 2.3).

2.1 Datasets

FOLIO is an NLI-through-FOL benchmark: each instance contains a multi-sentence story (premises) and a conclusion, both in NL and FOL, paired with a NLI label (TRUE/FALSE/UNKNOWN) indicating whether the conclusion is entailed by, contradicted by, or independent of the premises. The dataset is split randomly into training, validation, and test sets ($\sim$1360/275/275 instances).[2] FOLIO is the dominant benchmark for NL-to-FOL translation (Yang et al., 2024b; Brunello et al., 2026b) and neurosymbolic reasoning (Olausson et al., 2023; Ryu et al., 2025). We verify its validation split (FOLIO_VALIDATION).

MALLS is a large-scale autoformalization dataset of NL–FOL pairs synthetically generated by GPT-4, commonly used for training autoformalizers (Vossel et al., 2025; Thatikonda et al., 2024). The full dataset comprises $\sim$28K instances; 1000 were declared human-checked and reserved as a test set. We consider the first 100 of these (MALLS_TEST).[3]

We also include 213 instances from **GGC** (Barker-Plummer et al., 2011b) as a control, to verify that our curation framework does not introduce noise into already-correct annotations. GGC contains students' FOL submissions for *Tarski's World* exercises from *Language, Proof and Logic*

[2] Errors found in the validation split may hence apply to the other splits as well.

[3] The MALLS test set provides no metadata about the ordering of its instances, and the original paper does not document any explicit ordering criterion. In the absence of such information, no sampling strategy is demonstrably more representative than any other; we take the first 100 instances, considering it a transparent and reproducible choice.

Quantity	FOLIO	MALLS	GGC
Reviewed Dataset Size	275	100	213
Incorrect FOL sentences (# / %)	117/42.5%	42/42%	0/0.0%
– among which syntactically incorrect (# / %)	29/10.5%	3/3%	0/0.0%
– among which semantically incorrect (# / %)	88/32.0%	39/39%	0/0.0%
Ambiguous NL sentences (# / %)	49/17.8%	51/51%	18/8.4%
Ambiguous NL and incorrect FOL (# / %)	37/13.5%	19/19%	0/0.0%

Table 1: Errors and ambiguities identified during human curation. *Incorrect FOL sentences* counts formalizations that are wrong in the original datasets; subcategories are mutually exclusive.

(Barker-Plummer et al., 2011a), validated by instructors via an automatic tool, providing strong guarantees of syntactic and semantic correctness.[4]

2.2 Quality Analysis

Prior works (Olausson et al., 2023; Brunello et al., 2026b) have noted the presence of errors in FOLIO and MALLS; this paper aims to quantify their impact on evaluation outcomes.

In both datasets we identified both *translation errors* and *ambiguous NL sentences*. Both phenomena may penalize a model that produces correct formalizations: translation errors cause a valid formula to be scored against an incorrect reference, while ambiguous sentences allow multiple legitimate interpretations, only one of which is encoded in the ground truth. GGC showed no significant errors, with only a small portion of ambiguous sentences ($\sim$8.5%).

The original FOLIO_VALIDATION also contains *NLI labels inconsistent* with the provided formalizations; we corrected 17 such labels ($\sim$8.4% of the dataset) and re-verified them against NL reasoning.

In the following, we detail the annotation protocol, the issues identified during curation and the choices made to address them.

Annotation Protocol. The curation was conducted by two experienced reviewers—a PhD student and a research fellow, both with a mathematical background including coursework in logic. For each dataset, instances were split equally between the two annotators: each first reviewed their own half independently, and then cross-checked the half reviewed by the other. At this stage, the inter-annotator disagreement rate on the *correctness* of the original dataset formulas (i.e., cases where one annotator judged a formalization correct and the other did not) was 9.1% for FOLIO and 7% for MALLS. The disagreement rate on *ambiguity* (i.e., cases where one annotator judged a sentence ambiguous and the other did not) was 10.2% for FOLIO and 13% for MALLS.[5] All disagreements were subsequently resolved through discussion until consensus was reached.

Ontology Issues. Since no dataset provides an explicit ontology, we extract the signature—predicates, arities, and constants—automatically from the FOL formula and then we use an LLM to assign natural language meanings to each symbol, verifying then their consistency manually.

We encountered two distinct types of ontological issues. The first is *inconsistencies*: when a symbol appeared with conflicting arities or meanings across the premises and conclusion of the same story, or within the same formula, we manually corrected the ontology to restore consistency. In MALLS_TEST, we frequently found compound relational symbols (e.g., `WorksInNewsIndustryAndReportsOnEvents`) that obscure the logical structure. In such cases, we decomposed the predicate into its constituent parts (e.g., `WorksInNewsIndustry` and `ReportsOnEvents`) to recover a compositional ontology.

The second type is *ontological looseness*: unconventional but internally consistent logical conventions applied uniformly throughout a formalization. A concrete example is the following: the NL expression *school events* is encoded as the constant `schoolEvent`, even though FOL constants canonically denote specific individuals rather than classes. A more standard rendering would use a predicate `SchoolEvent(x)` ranging over individuals. This pattern recurs across several FOLIO stories.

Unlike the first type, cases of ontological looseness do not compromise the validity of the reasoning. Since ontology design admits genuine degrees of freedom, there is no principled basis for preferring a finer or coarser granularity in the absence of additional contextual information; therefore, in-

[4]GGC is not publicly available but is shared by the authors upon request.

[5]Percentages are computed over the total number of instances processed.

stances exhibiting these features are *not* regarded as errors in the dataset. In some cases, for the sake of adhering to standard conventions, we nonetheless rename constants or predicates, or reify concepts; however, as already noted, this "stylistic choice" does not affect the error count reported in Table 1.

Translation Errors. We identified recurring errors that compromise the correctness of the ground truth. *Syntactic errors* (parenthesis mismatches, symbol typos, ontology misuse, and free variables) affect 10.5% of instances in FOLIO and 3% in MALLS. The remaining are *semantic errors*, where a syntactically valid formula misrepresents the NL meaning: these are more prevalent (∼32% in FOLIO and ∼39% in MALLS) and harder to detect automatically. They include quantifier scope errors, missing information explicitly stated in the NL sentence, incorrect entity relativization, and broader logical structure failures.

A summary of these findings is reported in Table 1. Appendix B discusses how to reconcile these findings with seemingly contradictory LLM logical translation performance rates reported in the literature. The curated dataset releases include corrected ontologies (see Appendix C), and corrected formalizations together with explanations of each correction (see Appendix D, E, and F for details).

Ambiguities. The inherent ambiguity of NL sentences poses a fundamental challenge for NL-to-FOL formalization (Yadav et al., 2021): when acknowledged, it is typically addressed by requiring that sentences explicitly state all background information necessary for their interpretation, or that vague terms be avoided. Since no prior work provides systematic criteria for identifying ambiguity in NL-to-FOL benchmarks, we adopt an operative definition and annotate ambiguous instances explicitly during the human curation phase. We regard a sentence as ambiguous when, given a fixed ontology, multiple non-equivalent formalizations are defensible.

Ambiguity has direct consequences on evaluation: because all NL-to-FOL benchmarks supply a single reference formula, equivalence-based scoring penalizes a model when it proposes correct alternative formalizations.

Explicit ambiguity annotation thus serves a dual purpose: it enables downstream analyses to account for this confound, and it fosters a cleaner assessment of model capabilities. For a qualitative classification of the ambiguity types encountered across datasets, see Appendix G.

FOLIO's NLI Labels Inconsistencies. As already discussed, FOLIO is also an NLI dataset. For each set of premises $\{p_1, \ldots, p_n\}$ and conclusion c, there is also a label $l \in \{\text{TRUE}, \text{FALSE}, \text{UNKNOWN}\}$ that reflects if c is entailed by $p_1, \ldots, p_n$ (TRUE), is in contradiction with them (FALSE) or its truth value can't be deductively established (UNKNOWN).

FOLIO provides formalizations $\varphi_1, \ldots, \varphi_n$ of the story premises and ψ of the conclusion. These can be used to check whether the label l assigned to a pair premises-conclusion is correct: indeed, the entailment should be preserved in FOL, so the value of l should depend on whether from $\varphi_1, \ldots, \varphi_n$, one could derive ψ, or its negation, or neither of them.

Using the original FOL sentences, the label provided by the author is different from the one derived from the Z3 automatic solver (De Moura and Bjørner, 2008) 18% of times: this is largely motivated by the great amount of errors present in the FOL formulas, and partially (2%) by the fact that the inference works only assuming some background knowledge that is not explicitated in the NL sentence and it is not known if it can be assumed (hence formalized) or not.

Our curation (i) corrects the NLI labels (8.4% of the entire dataset), (ii) makes implicit background knowledge explicit when necessary for the entailment (through the `NLI_requires_more_info` tag), (iii) re-verifies the consistence between the new labels and the curated formalizations with the Z3 solver.

2.3 LLM Re-evaluation and Performance Shifts

We re-evaluate a selection of recent LLMs—Gemma 4 31B-it (Google, 2024), Qwen3-30B-A3B (Yang et al., 2025), and GPT-4o-mini (OpenAI, 2024)—on the NL-to-FOL translation task using Chain-of-Thought prompting with the target ontology provided in context.

Models are evaluated on both the original and curated datasets: given a NL sentence p and an ontology Ω, we ask the model to formalize p into a formula φ adhering to the ontology. We check (binary pass/fail) whether φ is logically equivalent to the reference formula: accuracy is the fraction of instances that are equivalent to the reference. In the original (uncurated) setting, the model receives the

Translation Accuracy [%]

Model	FOLIO						MALLS					
	All Instances			Unambiguous Only			All Instances			Unambiguous Only		
	Orig.	Cur.	Δ	Orig.	Cur.	Δ	Orig.	Cur.	Δ	Orig.	Cur.	Δ
Gemma 4 31B-it	50.9	**74.2**	**+23.3**	59.7	**82.7**	**+23.0**	66.0	**86.0**	**+20.0**	61.2	**93.9**	**+32.7**
Qwen3-30B-A3B	47.5	58.5	**+11.0**	55.2	65.0	**+9.8**	52.1	68.0	**+15.9**	53.3	65.3	**+12.0**
GPT-4o-mini	48.4	61.5	**+13.1**	57.1	69.0	**+11.9**	49.0	68.0	**+19.0**	40.8	65.3	**+24.5**

Table 2: Translation accuracy (%) on FOLIO_VALIDATION and MALLS_TEST under *Original* (Orig.) and *Curated* (Cur.) ground truth for all instances and the unambiguous subset. Δ = (Curated – Original); all values are positive, confirming that curation consistently raises measured accuracy.

original ontology and is scored against the original dataset formula as reference; in the curated setting, it receives our curated ontology and is scored against our curated formula (see Appendix H for details). As shown in Table 2, curation consistently raises measured accuracy across all models, with gains of +11 to +23 percentage points on FOLIO and +16 to +20 on MALLS. The largest improvements are exhibited by Gemma 4, the strongest model, suggesting that annotation noise disproportionately penalizes models capable of producing valid but non-reference formalizations. We also report results restricted to unambiguous instances, which offer a clear illustration of the impact of ambiguity on autoformalization datasets and provide a cleaner assessment of model capability that is not confounded by inherently ambiguous NL sentences: with respect to the total dataset, if we restrict on the unambiguous subset, Gemma 4 performs +8.5 and +7.9 percentage points better on FOLIO and MALLS, respectively.

3 LLM-assisted Human Oversight

Manual inspection is the most reliable way to ensure ground-truth quality, but does not scale: costs (money and time) grow linearly with dataset size, making exhaustive review impractical for large corpora. Delegating verification to automation is equally unsatisfactory, since assessing the alignment between a NL sentence and its formal representation remains an open problem (Brunello et al., 2026a; Mensfelt et al., 2026). We therefore investigate a middle ground: using LLMs as judges (Zheng et al., 2023) to build a first-pass filter that concentrates human attention on instances most likely to contain errors. To this end, in this section we describe the *Verdict-and-Refinement Task*, two pipeline designs that instantiate it, and the resulting prioritization strategy for LLM-assisted dataset curation.

3.1 Verdict-and-Refinement Task

We define the *Verdict-and-Refinement* task (V&R) as follows: given a NL sentence p, a candidate FOL formula φ, and an ontology Ω (predicates, constants, and their associated meanings), the goal is to produce (i) a *verdict* $v \in \{\mathsf{yes}, \mathsf{no}, \mathsf{?}\}$ and (ii) a *refinement* FOL formula ψ. Specifically:

- $v = \mathsf{yes}$: φ correctly formalizes p in the given ontology Ω; no refinement is needed;
- $v = \mathsf{no}$: φ incorrectly formalizes p under Ω; the refinement ψ should provide a correct formalization of p;
- $v = \mathsf{?}$: the model is uncertain, or p is ambiguous, i.e., it admits multiple non-equivalent yet defensible formalizations under Ω: in the latter case, ψ should capture the most natural interpretation of p.

The three-way verdict reflects a deliberate design choice, driven primarily by the treatment of ambiguity. A binary yes/no decision would force the model to either reject legitimate alternative interpretations or arbitrarily privilege one reading as canonical; the ? label instead explicitly accommodates ambiguity while also allowing abstention under uncertainty.

3.2 Pipelines

Considering a dataset whose instances are triplets (p, Ω, φ) as in Section 3.1, we embed the V&R task within two distinct pipelines, illustrated in Figure 1.

- **Pipeline 1: Direct V&R.** The LLM receives the triplet (p, Ω, φ) and produces a verdict v and, optionally, a refinement ψ when φ is judged incorrect or ambiguous. The ultimately proposed formula is $\hat{\psi} = \varphi$ if $v = \mathsf{yes}$, and $\hat{\psi} = \psi$ otherwise.
- **Pipeline 2: Re-generation and V&R.** The original formula φ is discarded. The LLM

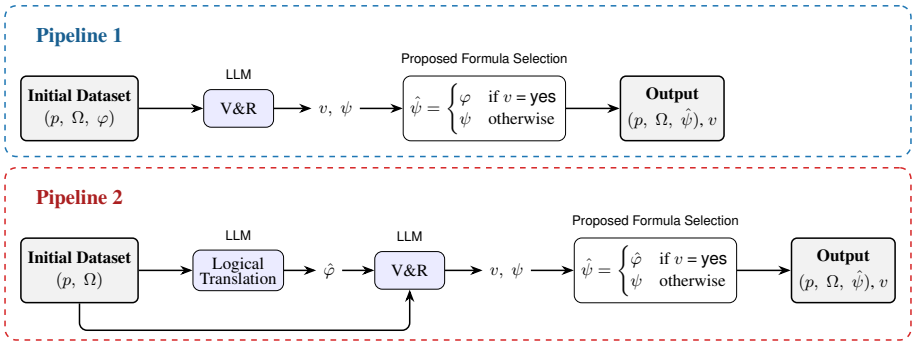


Figure 1: The two pipelines. Each starts from the *Initial Dataset* containing triplets (p, Ω, φ), and produces an output with the *Formalization Proposal* $(p, \Omega, \hat{\psi})$ and the *verdict* v. *Pipeline 1* judges the original formula φ directly. *Pipeline 2* first re-generates a candidate $\hat{\varphi}$ from p and Ω alone, then judges it.

first translates p under Ω into a candidate $\hat{\varphi}$, which is then submitted to the same V&R step as in Pipeline 1. The finally proposed formula is $\hat{\psi} = \hat{\varphi}$ if $v = \textsf{yes}$, and $\hat{\psi} = \psi$ otherwise.

The two pipelines embody different assumptions about the original formula φ. Pipeline 1 treats it as a reliable anchor: refining an existing formula may be more effective than generating one from scratch. Pipeline 2 treats it as a potential liability: when φ is unreliable or difficult for the model to evaluate, retaining it may mislead the V&R step, and a freshly generated candidate can be a safer starting point.

Each pipeline processes the original dataset and produces an output comprising two elements for each instance: (i) the *formalization proposal* consisting of triplets $(p, \Omega, \hat{\psi})$, where $\hat{\psi}$ is the pipeline's proposed formula, and (ii) the *verdict* v emitted during the V&R step, referring to the original formula φ or to the LLM-generated candidate $\hat{\varphi}$, respectively in Pipeline 1 and Pipeline 2. Despite this difference in reference, in both cases the verdict carries information about the reliability of $\hat{\psi}$: a yes verdict signals that $\hat{\psi}$ derives from a formula that was deemed correct; no indicates it refines a formula deemed incorrect; and ? flags either ambiguity in p or model uncertainty.

3.3 Prioritization Order

Assume a pipeline is run on all instances of a dataset, producing outputs $(p, \Omega, \hat{\psi})$, each paired with a verdict v. The verdicts v can be used to prioritize manual inspection, by inspecting first the instances in a certain verdict class, followed by others from another class, and so on: we call this a *prioritization order*. Intuitively, for example, a yes verdict means formula $\hat{\psi}$ was considered correct by both the generator (Pipeline 1: dataset creator; Pipeline 2: LLM) and the V&R judge, making it the least likely to contain errors, and hence the last to review. In Section 5, we show how to find an adequate prioritization order.

4 Experimental Setup

To evaluate our proposed LLM-assisted oversight approach, we require ground-truth annotations for each instance. We therefore assume every original triplet (p, Ω, φ) is accompanied by (i) a gold verdict $v_{\text{gold}} \in \{\textsf{yes}, \textsf{no}, \textsf{?}\}$ assessing the correctness of φ or ambiguity of p, and (ii) a gold formalization φ_{gold} of p under Ω.

Specifically, for each instance (p, φ) in the original FOLIO_VALIDATION and MALLS_TEST, we derive the ontology Ω from φ and construct the curated ground-truth φ_{gold}; given this, we derive the verdict v_{gold} marking as yes, no, and ? the instances for which $\varphi \equiv \varphi_{\text{gold}}$, for which $\varphi \not\equiv \varphi_{\text{gold}}$, and for which p is ambiguous. This is exactly the process used in Section 3 for curating the datasets. We also perform our evaluation on GGC to measure the possible degradation introduced by the pipelines when applied to already-correct data. Since GGC annotations are verified correct, $v_{\text{gold}} = \textsf{yes}$ and $\varphi_{\text{gold}} = \varphi$ for all instances, except for the few ambiguous sentences (where $v_{\text{gold}} = \textsf{?}$).

Simulating human review. Our primary interest is in measuring how effectively each proposed pipeline, when coupled with a prioritization order, concentrates errors at the top of the inspection queue, so that a human annotator achieves high accuracy while reviewing as few instances as possible. We model this by simulating a perfect annotator who, upon inspecting an instance, immediately cor-

rects it. We track *accuracy*—the fraction of correct instances in the dataset—as a function of the fraction of instances reviewed by the human, $h \in [0, 1]$; we refer to this as the *accuracy–human effort curve*.

Two baselines provide reference curves against which to compare the prioritization effectiveness of the pipelines: (i) **Black Baseline (original dataset)**—the annotator reviews instances from the original dataset in random order. Accuracy starts at the original dataset accuracy and increases linearly to 100% as h reaches 1; (ii) **Green Baseline (LLM regeneration)**—the annotator starts from a synthetic dataset produced in the Logical Translation phase of Pipeline 2 and reviews instances in random order. Accuracy starts at the regenerated dataset accuracy and increases linearly to 100% as h reaches 1.

Metrics. As we mentioned, each strategy corresponds to an accuracy–human effort curve, with steeper early rises indicating more effective prioritization. Our primary metric is the *Area Under the Curve* (AUC; higher is better). Since AUC alone is not easily interpretable, we complement it with T_α, the minimum fraction of instances to review to reach $\alpha\%$ accuracy (lower is better). We report T_{90} and T_{95}.

Models and prompting. We run all pipelines with three models: GPT-4o-mini, Qwen3-30B-A3B, and Gemma 4 31B-it. Following best practices for LLM evaluation (Mizrahi et al., 2024; Sclar et al., 2024), and since our goal is not to identify a single optimal prompt but rather to assess whether the capabilities required for our application can be consistently elicited, for each model we considered four prompting strategies varying in reasoning approach (standard CoT (B1), few-shot (B2), few-shot + CoT (B3), and meta-prompting (B4)) and four prompt variants (pv1–pv4) differing in their degree of skepticism toward the input formula and level of instructional detail, yielding a total of 16 prompting combinations. Full details are in Appendix H.

5 Results and Discussion

For each of the two pipelines, in Appendix I, we measure accuracy within each verdict class and find that it increases monotonically from ? to no to yes. We thus adopt the following prioritization order for both of them: (i) malformed outputs, (ii) ?, (iii) no, (iv) yes. All results below are reported under this prioritization.[6]

Figure 2 compares the prioritization strategies across datasets and models. In terms of AUC, Pipeline 1 (blue) outperforms Pipeline 2 (red), which in turn outperforms the Green Baseline (green). This ordering holds across all metrics (T_{90}, T_{95}) and is statistically significant (Appendix K).[7] Among the models evaluated, Gemma performs best (Appendix L).

Given the witnessed results, we focus on Gemma with Pipeline 1 for the main analysis. Among the prompting strategies, B4 and variants pv2 and pv4 perform significantly worse than the remaining configurations; the others (B1, B2, B3 with pv1 and pv3) are statistically indistinguishable, indicating that Pipeline 1 is robust to prompting choices within this set (Appendix M). See Appendix N for the results for all models and datasets.

Figure 3 plots the *accuracy–human effort curves*, showing the full range of Pipeline 1 performance across the retained prompting configurations (blue band) and the best-AUC configuration for Pipeline 2. The ordering among Green Baseline, Pipeline 2, and Pipeline 1 is preserved across all datasets and metrics.

The steep initial slopes confirm that the above prioritization order effectively front-loads error-prone instances. The practical gains are substantial: on FOLIO, 90% accuracy is reached after reviewing only 20% of instances— a third of the effort required by the Green Baseline (61%) and less than a third of that required by the Black Baseline (76%). On MALLS, the reduction is even more pronounced: 5% suffices against 29% for the Green Baseline (a $> 5\times$ reduction) and against 76% for the Black Baseline (a $15\times$ reduction). Comparable gains hold for T_{95}.

Furthermore, in both datasets, instances assigned a yes verdict constitute around half of the data; accuracy restricted to such a class is above 97%. This opens the possibility of treating such a class as a ready-to-use curated training set; however, doing so halves the dataset and may disproportionately discard the most challenging instances—a trade-off that practitioners should weigh carefully.

Finally, when applied to the already 100% cor-

[6] Note that other prioritization strategies can, in principle, be adopted, including strategies that do not rely on the verdict produced by the V&R task. See Appendix J for an alternative proposal of this kind.

[7] For simplicity, these figures omit the Black Baseline, which is based on the original dataset; its (bad) performance is addressed below.

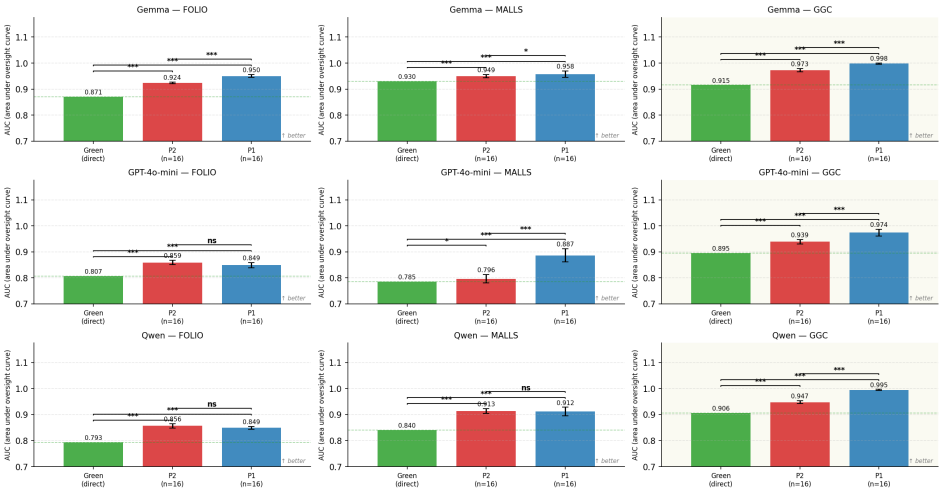


Figure 2: Pipelines comparison across models (horizontally) and datasets (vertically) according to the AUC (Area Under the Curve) metric. Asterisks denote statistical significance (*** = $p < 0.001$, ** = $p < 0.01$, * = $p < 0.05$) under the one-sided paired Wilcoxon signed-rank test. ns: non-significant.

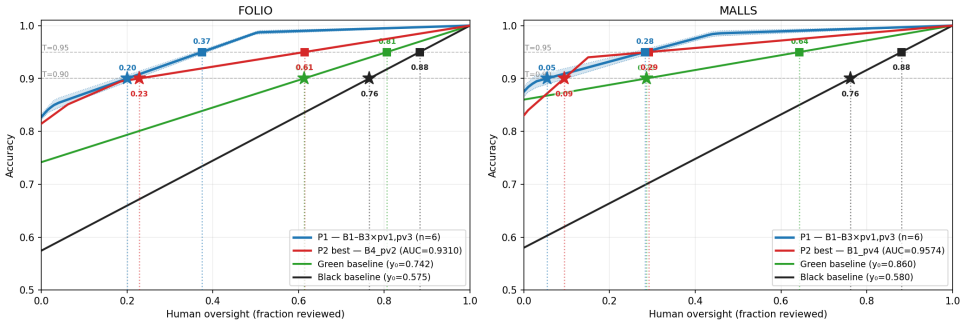


Figure 3: *Accuracy-human effort* curve. Each plot shows Gemma's performance on FOLIO_VALIDATION (left) and MALLS_TEST (right). The blue band represents Pipeline 1 (min/max and average across the six prompting combinations {B1, B2, B3}×{pv1, pv3}); the red curve shows the best-AUC configuration of Pipeline 2; the black and green line represents respectively the Black and Green Baseline.

rect dataset GGC, all three models with Pipeline 1 produce a limited degradation of $\leq 5\%$ (see Appendix N).

6 Conclusions and Future Works

This work addresses systematic annotation errors in two widely used NL-to-FOL datasets, FOLIO_VALIDATION and MALLS_TEST, uncovering error rates of 42.5% and 42% respectively. These errors measurably distort LLM evaluation, with performance estimates shifting by up to 23 points once corrected annotations are used. To address the infeasibility of exhaustive human oversight, we propose an LLM-assisted framework that concentrates human effort on the instances most likely to contain errors. The best pipeline configurations allow a human annotator to reach 90% accuracy after reviewing only 20% of FOLIO_VALIDATION and 5% of MALLS_TEST.

Future investigation includes: (i) as a natural extension, applying the pipeline to the full training splits of FOLIO and MALLS, where exhaustive manual review is prohibitive but systematic quality improvements would have broad impact; (ii) increasing pipeline autonomy through iterative refinement loops and automated ontology construction would reduce the remaining manual effort, (iii) introducing a dedicated pre-processing stage to detect and resolve linguistic ambiguity prior to formalization is a promising direction toward a more robust curation process.

7 Limitations

Dataset curation is inherently delicate and error-prone. Despite the strategies adopted to remain as objective as possible, curation remains an inher-

ently subjective process, and residual annotation mistakes may persist. We therefore expect that the exact error and ambiguity counts reported here could shift slightly upon further scrutiny. However, this does not undermine our core message: these datasets contain a substantial number of annotation errors and ambiguities that meaningfully compromise the reliability of LLM evaluation on them. Annotators corrected instances, without access to the outputs of the models evaluated in Section 2.3 and Section 5, ruling out any risk of bias toward the reasoning patterns of a specific tested model. A further limitation in this regard concerns ambiguity: the criteria used to flag ambiguous sentences are not standardized by prior work, and alternative strategies may be equally defensible.

Finally, the proposed prioritization strategy is shown to work well for the models and datasets tested, and it relies on the empirical evidence that LLMs rarely flag correct formalizations as incorrect. However, when applying the strategy to a new dataset or a new model, we advise validating this underlying assumption in the new setting as well, using a small calibration set.

References

Dave Barker-Plummer, Richard J. Cox, and Robert Dale. 2009. Dimensions of difficulty in translating natural language into first-order logic. In *Educational Data Mining*.

Dave Barker-Plummer, Richard J. Cox, and Robert Dale. 2011a. *Language, Proof, and Logic*, 2nd edition. Center for the Study of Language and Information/SRI.

Dave Barker-Plummer, Richard J. Cox, and Robert Dale. 2011b. Student translations of natural language into logic: The Grade Grinder corpus release 1.0. In *EDM*, pages 51–60.

Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, and 12 others. 2020. Language models are few-shot learners. *Preprint*, arXiv:2005.14165.

Andrea Brunello, Cristian Curaba, Luca Geatti, Michele Mignani, Angelo Montanari, and Nicola Saccomanno. 2026a. Natural language to first-order logic llm-based autoformalization. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics. To appear.

Andrea Brunello, Luca Geatti, Michele Mignani, Angelo Montanari, and Nicola Saccomanno. 2026b. Do LLMs really struggle at NL-FOL translation? revealing their strengths via a novel benchmarking strategy. In *Proceedings of the 2026 Annual meeting of the Association for the Advancement of Artificial Intelligence (AAAI)*, volume 40, pages 30094–30103.

David "davidad" Dalrymple, Joar Skalse, Yoshua Bengio, Stuart Russel, Max Tegmark, Sanjit Seshia, Steve Omohundro, Christian Szegedy, Ben Goldhaber, Nora Ammann, Alessandro Abate, Joe Halpern, Clark Barrett, Ding Zhao, Tan Zhi-Xuan, Jeannette Wing, and Joshua Tenenbaum. 2024. Towards guaranteed safe AI: A framework for ensuring robust and reliable AI systems. *CoRR*, abs/2405.06624.

Leonardo De Moura and Nikolaj Bjørner. 2008. Z3: An efficient smt solver. In *International conference on Tools and Algorithms for the Construction and Analysis of Systems (ITACAS)*, pages 337–340. Springer.

Google. 2024. Gemma 4. `https://ai.google.dev/gemma/docs/core`. Accessed: 2026-05-04.

Simeng Han, Hailey Schoelkopf, Yilun Zhao, Zhenting Qi, Martin Riddell, Wenfei Zhou, James Coady, David Peng, Yujie Qiao, Luke Benson, Lucy Sun, Alexander Wardle-Solano, Hannah Szabó, Ekaterina Zubova, Matthew Burtell, Jonathan Fan, Yixin Liu, Brian Wong, Malcolm Sailor, and 16 others. 2024.

FOLIO: natural language reasoning with first-order logic. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 22017–22031. Association for Computational Linguistics.

Agnieszka Mensfelt, David Tena Cucala, Santiago Franco, Angeliki Koutsoukou-Argyraki, Vince Trencsenyi, and Kostas Stathis. 2026. Towards a common framework for autoformalization. In *Proceedings of the 2026 Annual meeting of the Association for the Advancement of Artificial Intelligence (AAAI)*, pages 40971–40980. AAAI Press.

Moran Mizrahi, Guy Kaplan, Dan Malkin, Rotem Dror, Dafna Shahaf, and Gabriel Stanovsky. 2024. State of what art? a call for multi-prompt LLM evaluation. *Transactions of the Association for Computational Linguistics (TACL)*, 12:933–949.

Theo Olausson, Alex Gu, Ben Lipkin, Cedegao Zhang, Armando Solar-Lezama, Joshua Tenenbaum, and Roger Levy. 2023. LINC: A neurosymbolic approach for logical reasoning by combining language models with first-order logic provers. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 5153–5176. Association for Computational Linguistics.

OpenAI. 2024. Gpt-4o-mini. `https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/`. Accessed: 2026-05-04.

Liangming Pan, Alon Albalak, Xinyi Wang, and William Yang Wang. 2023. Logic-lm: Empowering large language models with symbolic solvers for faithful logical reasoning. In *Findings of the Association for Computational Linguistics: EMNLP 2023, Singapore, December 6-10, 2023*, pages 3806–3824. Association for Computational Linguistics.

Hyun Ryu, Gyeongman Kim, Hyemin S. S. Lee, and Eunho Yang. 2025. Divide and translate: Compositional first-order logic translation and verification for complex logical reasoning. In *Proceedings of the 2025 International Conference on Learning Representations (ICLR)*.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2024. Quantifying language models' sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting. In *Proceedings of the 2024 International Conference on Learning Representations (ICLR)*. OpenReview.net.

Hrituraj Singh, Milan Aggarwal, and Balaji Krishnamurthy. 2020. Exploring neural models for parsing natural language into first-order logic. *CoRR*, abs/2002.06544.

Ramya Keerthy Thatikonda, Jiuzhou Han, Wray Buntine, and Ehsan Shareghi. 2024. Strategies for improving NL-to-FOL translation with LLMs: Data generation, incremental fine-tuning, and verification. *CoRR*, abs/2409.16461.

Felix Vossel, Till Mossakowski, and Björn Gehrke. 2025. Advancing natural language formalization to first order logic with fine-tuned LLMs. *CoRR*, abs/2509.22338.

Jun Wang, David Smith Sundarsingh, Jyotirmoy V. Deshmukh, and Yiannis Kantaros. 2025. ConformalNL2LTL: Translating natural language instructions into temporal logic formulas with conformal correctness guarantees. *CoRR*, abs/2504.21022.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2023. Chain-of-thought prompting elicits reasoning in large language models. *Preprint*, arXiv:2201.11903.

Apurwa Yadav, Aarshil Patel, and Manan Shah. 2021. A comprehensive review on resolving ambiguities in natural language processing. *AI Open*, 2:85–92.

An Yang and 1 others. 2025. Qwen3 technical report. *CoRR*, abs/2505.09388.

Yuan Yang, Siheng Xiong, Ali Payani, Ehsan Shareghi, and Faramarz Fekri. 2024a. Harnessing the power of large language models for natural language to first-order logic translation. In *Proceedings of the 2024 Annual meeting of the Association of Computational Linguistic (ACL)*, pages 6942–6959. Association for Computational Linguistics.

Yuan Yang, Siheng Xiong, Ali Payani, Ehsan Sharenghi, and Faramarz Fekri. 2024b. Harnessing the power of large language models for natural language to first-order logic translation. In *Proceedings of the 2024 Annual meeting of the Association of Computational Linguistic (ACL)*, pages 6942–6959. Association for Computational Linguistics.

Xi Ye, Qiaochu Chen, Isil Dillig, and Greg Durrett. 2023. SatLM: Satisfiability-aided language models using declarative prompting. In *Proceedings of the 2023 Annual Conference on Neural Information Processing Systems (NeurIPS)*.

Yifan Zhang, Yang Yuan, and Andrew Chi-Chih Yao. 2024. Meta prompting for AI systems. *Preprint*, arXiv:2311.11482.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Proceedings of the 2023 Annual Conference on Neural Information Processing Systems (NeurIPS)*.

A Autoformalization in FOL: Background

First-Order Logic. A *signature* $\sigma = (\mathcal{V}, \mathcal{C}, \mathcal{F}, \mathcal{R})$ consists of a countably infinite set of variables $\mathcal{V}$, a set of constants $\mathcal{C}$, a set of function symbols $\mathcal{F}$, and a set of relation/predicate

symbols $\mathcal{R}$ (each with an associated arity), where all four sets are pairwise disjoint. *Terms* are built inductively: variables and constants are base terms; if $t_1, \ldots, t_n$ are terms and $f \in \mathcal{F}$ has arity n, then $f(t_1, \ldots, t_n)$ is a term. *FOL formulas* over σ are defined inductively by:

$$\varphi \;:= r(t_1, \ldots, t_n) \mid \neg\varphi \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid \varphi \oplus \varphi \mid \varphi \to \varphi \mid \varphi \leftrightarrow \varphi \mid \exists x\, \varphi \mid \forall x\, \varphi,$$

where $r \in \mathcal{R}$ with arity n, the t_i are terms, and $\neg, \wedge, \vee, \oplus, \to, \leftrightarrow, \exists, \forall$ denote negation, conjunction, inclusive disjunction, exclusive disjunction, implication, equivalence, and the existential and universal quantifiers respectively.

Operator precedence. When parentheses are omitted, the logical operators above are assumed to bind in the following order (from highest to lowest precedence):

$$\{\neg, \exists, \forall\} \;\succ\; \wedge \;\succ\; \vee \;\succ\; \to \succ \leftrightarrow \succ \oplus.$$

For example, $\neg A \wedge B \to C \vee D$ is parsed as $((\neg A) \wedge B) \to (C \vee D)$. Parentheses are required explicitly whenever the intended scoping departs from this convention.

Ontology. Evaluating whether a FOL formula faithfully represents a natural language sentence requires an interpretation of the symbols in the formula. We use the term *ontology* to refer to the signature enriched with the intended NL meaning of each symbol (e.g., $\text{Tet}(x)$ means "x is a tetrahedron"). We denote this usually with the symbol Ω.

The autoformalization task. Autoformalization is the task of producing a FOL representation that symbolically captures the semantic meaning of a NL phrase p. Following (Brunello et al., 2026b), the task decomposes along two dimensions:

- **Ontology Extraction (OE):** identifying a suitable signature for the formalization, along with the interpretation of the symbols in it.
- **Logical Translation (LT):** translating p into a FOL formula using the symbols and meanings provided by a given ontology.

This decomposition arises from the fact that determining whether a given formula is equivalent to another is only possible when the two formulas share the same signature. Therefore, in this paper, whenever we ask a model to produce a formalization, we also provide the signature as part of the input (LT task). This allows us to reliably evaluate performance on the formalization task automatically.

As noted in (Brunello et al., 2026b), without a labeled signature it is also impossible to manually assess whether a formula conveys the same meaning as a natural language sentence, since the interpretation of the symbols used in the formula would remain unclear.

For a more complete definition of the FOL-autoformalization task and for a survey of approaches, datasets, and open chellenges in NL-to-FOL formalization, see (Brunello et al., 2026a).

B Dataset Error Rates are Coherent with Recent Literature

At first glance, the high percentage of incorrect formalizations we identify in the FOLIO_VALIDATION dataset (42.5%) may seem to contradict recent findings in the literature. For instance, (Brunello et al., 2026b) demonstrated that state-of-the-art models like o3-mini can achieve approximately 80% accuracy in NL to FOL translation tasks on the train split of FOLIO dataset.

Sentence-Level vs. Mixed-Instance Evaluation. The discrepancy stems from the fundamental unit of analysis. Evaluations that report $\sim$80% accuracy assess *isolated sentences* translated into standalone FOL formulas. In contrast, our quality assessment of FOLIO evaluates the dataset according to its actual NLI structure.

The FOLIO_VALIDATION split contains 275 total instances, divided into two distinct types:

1. **Stories (73 instances):** The concatenated premises of an NLI problem, averaging roughly 6 natural language sentences each.
2. **Conclusions (202 instances):** Single-sentence logical conclusions.

In the context of automated reasoning, a story's formalization is only correct if *every single component sentence* is flawlessly translated and ontologically consistent. The reason behind the choice to consider the story as a whole is that some information is not explicit in specific sentences and derives from the context posed by the premises in the story: thus, if we have in mind general applications of the autoformalization task, ours seems the most natural choice.

Mathematical Justification. We can demonstrate that our 42.5% instance-level error rate is entirely consistent with ∼80% sentence-level accuracy as reported in the literature.

The FOLIO_VALIDATION dataset contains 275 instances composed of 73 multi-sentence stories (averaging 6 sentences each) and 202 single-sentence conclusions, yielding approximately $73 \times 6 + 202 = 640$ total sentences. Supposing that any error done by the model in FOLIO stories are located in only one sentence, based on our error rate, we get $42.5\% \times 275 \sim 117$ incorrect sentences. A perfect annotator then would have produced an accuracy of:

$$\frac{640 - 117}{640} \sim 82\%$$

which is consistent with the ∼80% accuracy reported by (Brunello et al., 2026b) when evaluating state-of-the-art models on isolated FOLIO sentences.

We also notice that the dataset used for the evaluation in (Brunello et al., 2026b) is different from the one used here (FOLIO_TRAINING vs. FOLIO_VALIDATION) even if they are random splits coming from the same pool.

C Instance of The Ontology File

We report here an example of the entries of the ontology file given within the code. The following JSON is what can be found for the FOLIO instance `story_380`. Each entry in `Rel` corresponds to a predicate symbol extracted from the curated FOL formula, annotated with its arity (`arity`) and NL meanings (i.e., its NL interpretation (`pos_meaning`), and the meaning when the predicate is negated (`neg_meaning`)); argument positions are indicated by `{0}`, `{1}`, etc. Each entry in `Const` maps a logical constant name to its NL interpretation as it appears in the following example.

For FOLIO, each ontology is described by the `story_id`: for each premises in the multi-sentence story and for each conclusion related to the story, the formalization in FOL provided in the curated dataset adhere to the symbols here listes. For MALLS, each sentence comes with an own ontology: hence, we will have 73 entries for FOLIO and 100 for MALLS.


```
{
  "story_id": 380,
  "Rel": {
    "InThisClub":     { "arity": 1, "pos_meaning
    ": "{0} is in this club",
                                    "neg_meaning
    ": "{0} is not in this club" },
    "PerformOftenIn": { "arity": 2, "pos_meaning
    ": "{0} performs often in {1}",
                                    "neg_meaning
    ": "{0} does not perform often in {1}" },
    "Attend":         { "arity": 2, "pos_meaning
    ": "{0} attends {1}",
                                    "neg_meaning
    ": "{0} does not attend {1}" },
    "VeryEngagedWith":{ "arity": 2, "pos_meaning
    ": "{0} is very engaged with {1}",
                                    "neg_meaning
    ": "{0} is not very engaged with {1}" },
    "InActive":       { "arity": 1, "pos_meaning
    ": "{0} is inactive",
                                    "neg_meaning
    ": "{0} is not inactive" },
    "Disinterested":  { "arity": 1, "pos_meaning
    ": "{0} is disinterested",
                                    "neg_meaning
    ": "{0} is not disinterested" },
    "MemberOf":       { "arity": 2, "pos_meaning
    ": "{0} is a member of {1}",
                                    "neg_meaning
    ": "{0} is not a member of {1}" },
    "Chaperone":      { "arity": 2, "pos_meaning
    ": "{0} chaperones {1}",
                                    "neg_meaning
    ": "{0} does not chaperone {1}" },
    "Student":        { "arity": 1, "pos_meaning
    ": "{0} is a student",
                                    "neg_meaning
    ": "{0} is not a student" },
    "AttendSchool":   { "arity": 1, "pos_meaning
    ": "{0} attends school",
                                    "neg_meaning
    ": "{0} does not attend school" },
    "YoungChild":     { "arity": 1, "pos_meaning
    ": "{0} is a young child",
                                    "neg_meaning
    ": "{0} is not a young child" },
    "Teenager":       { "arity": 1, "pos_meaning
    ": "{0} is a teenager",
                                    "neg_meaning
    ": "{0} is not a teenager" },
    "WishToFurther":  { "arity": 2, "pos_meaning
    ": "{0} wishes to further {1}",
                                    "neg_meaning
    ": "{0} does not wish to further {1}" }
  },
  "Const": {
    "bonnie":                "Bonnie",
    "schoolTalentShow":      "the school talent
    show",
    "schoolEvent":           "school events",
    "community":             "the community",
    "highSchoolDance":       "high school dances
    ",
    "academicCareer":        "academic career
    and educational opportunities",
    "educationalOpportunity":"educational
    opportunities"
  }
}
```

D Released Data Format Of Human-Annotated Subsets

We release the human-annotated portions of FOLIO (275 instances) and MALLS (100 instances) as JSON records designed for reliable evaluations.

FOLIO: story records and conclusion records. We store each FOLIO instance as a *story record* (containing the full multi-sentence NL premises (story-level context) together with its FOL translation), or as a *conclusion record* (each record contains a conclusion in NL with its FOL counterpart). An example is reported in Appendix E.

Common fields. Both record types share:

- `id`: unique identifier (e.g., `story_471`, `concl_471_20`).[8]
- `NL_sentence`: the NL text.
- `FOL_sentence_old`: original formula from the source dataset.
- `FOL_sentence_new`: curated formula (equal to `FOL_sentence_old` if unchanged).
- `corrected`: whether the new FOL formula (`FOL_sentence_new`) is different from the original one (`FOL_sentence_old`).
- `is_originally_correct`: whether the original FOL formula (`FOL_sentence_old`) is considered correct or incorrect, when computing the errors of the dataset. Notice that there can be cases in which `corrected` is ýesánd also `is_originally_correct` also: in this case, it means that we are in the case of *ontological loseness* discussed in Section 3 and some usually adopted conventions (about namings or concepts reification) are applied to the original formulas.

Ambiguity metadata. If the NL admits multiple plausible readings, we set:

- `ambiguity`: boolean.
- `ambiguity_explanation`: categorical tag `some_plural`, `either_or`, `option_partition`, `sufficient_necessary_condition`, `determinative_the`, `existential_universal` when possible or otherwise an explanation of the ambiguity we detected.

Even when `ambiguity=true` we provide, in the `FOL_sentence` field, the most plausible alternative given the instructions detailed in Section Section 2.2.

Correction explanation. When `corrected=true`, we include:

- `correction_explanation`: short description of what was changed and why.

NLI labels (conclusion records only). Conclusion records additionally include:

- `label_old`: the original FOLIO NLI label.
- `label_old_Z3`: the NLI label derivable from the original formalizations
- `label`: our corrected label (coherent with `label_Z3`)
- `label_Z3`: the NLI label derivable from the new formalizations (plus eventually the information in the `NLI_requires_more_info`.
- `NLI_requires_more_info`: the FOL formula that represents the background knowledge that must be added to the story to ensure that the NLI label is consistent with the FOL entailment.

MALLS: item records. MALLS uses a single-item format, so each instance corresponds to a unique sentence always. Each record includes `id`, `NL_sentence`, `FOL_sentence_old`, `FOL_sentence_new`, `corrected`, and (when applicable) `ambiguity` with `ambiguity_explanation` as explained above. You can find an example in the Appendix F.

Ontology. Each FOLIO story with conclusions and MALLS instance is further accompanied by an *ontology* (signature enriched with LLM-generated glosses): see Appendix C.

[8] For the story records, the last number identifies the story ID in the original dataset. For the conclusion records, the first number identifies the story ID to which the conclusion is linked, while the second number identifies the row in which the conclusion appears in the original version of the dataset hosted on Hugging Face, available at https://huggingface.co/datasets/yale-nlp/FOLIO/viewer/default/validation.

E Example of Released FOLIO_VALIDATION_CURATED Dataset

Story record.


```
{
 {"id": "story_380",
 "NL_sentence": "People in this club who perform
     in school talent shows often attend and are
     very engaged with school events. People in
    this club either perform in school talent
    shows often or are inactive and
    disinterested community members. People in
    this club who chaperone high school dances
    are not students who attend the school. All
    people in this club who are inactive and
    disinterested members of their community
    chaperone high school dances. All young
    children and teenagers in this club who wish
     to further their academic careers and
    educational opportunities are students who
    attend the school. Bonnie is in this club
    and she either both attends and is very
    engaged with school events and is a student
    who attends the school or is not someone who
     both attends and is very engaged with
    school events and is not a student who
    attends the school.",
 "FOL_sentence_new": "∀x (InThisClub(x) ∧
    PerformOftenIn(x, schoolTalentShow) →
    Attend(x, schoolEvent) ∧ VeryEngagedWith(x,
     schoolEvent)) ∧ ∀x (InThisClub(x) → (
    PerformOftenIn(x, schoolTalentShow) ⊕ (
    InActive(x) ∧ Disinterested(x) ∧ MemberOf(x,
     community)))) ∧ ∀x (InThisClub(x) ∧
    Chaperone(x, highSchoolDance) → ¬(Student(
    x) ∧ AttendSchool(x))) ∧ ∀x (InThisClub(x)
    ∧ (InActive(x) ∧ Disinterested(x) ∧
    MemberOf(x, community)) → Chaperone(x,
    highSchoolDance)) ∧ ∀x (InThisClub(x) ∧ (
    YoungChild(x) ∨ Teenager(x)) ∧
    WishToFurther(x, academicCareer) ∧
    WishToFurther(x, educationalOpportunity) →
    (Student(x) ∧ AttendSchool(x))) ∧
    InThisClub(bonnie) ∧ ((Attend(bonnie,
    schoolEvent) ∧ VeryEngagedWith(bonnie,
    schoolEvent) ∧ Student(bonnie) ∧
    AttendSchool(bonnie)) ⊕ (¬(Attend(bonnie,
    schoolEvent) ∧ VeryEngagedWith(bonnie,
    schoolEvent)) ∧ ¬(Student(bonnie) ∧
    AttendSchool(bonnie))))",
 "corrected": "yes",
 "ambiguity": true,
 "ambiguity_explanation": "either_or",
 "correction_explanation": "1. Spelling:
    Corrected 'Studen' to 'Student' and 'bonne'
    to 'bonnie'. 2. Linkage: 'highSchoolDances'
    changed to 'highSchoolDance' in Premise 4.
    3. Missing Parenthesis & Operators: Fixed
    the missing opening parenthesis for the
    implication in Premise 5. Changed '
    YoungChildren' to 'YoungChild' and XOR '⊕'
    to inclusive OR '∨' in Premise 5 to match '
    young children and teenagers'. Added missing
     'educationalOpportunity' to WishToFurther
    to fully capture the NL. 4. Premise 6:
    Removed hallucinated outer parentheses added
     by the previous reviewer and properly
    bounded the variables to 'bonnie'.",
 "FOL_sentence_old": "∀x (InThisClub(x) ∧
    PerformOftenIn(x, schoolTalentShow) →
    Attend(x, schoolEvent) ∧ VeryEngagedWith(x,
     schoolEvent))\n∀x (InThisClub(x) →
    PerformOftenIn(x, schoolTalentShow) ⊕ (
    InActive(x) ∧ Disinterested(x) ∧ MemberOf(x,
     community)))\n∀x (InThisClub(x) ∧
    Chaperone(x, highSchoolDance) → ¬(Studen(x)
     ∧ AttendSchool(x)))\n∀x (InThisClub(x) ∧ (
    InActive(x) ∧ Disinterested(x) ∧ MemberOf(x,
     community)) → Chaperone(x,
    highSchoolDances))\n∀x (InThisClub(x) ∧ (
    YoungChildren(x) ⊕ Teenager(x)) ∧
    WishToFurther(x, academicCareer)) → Studen(
    x) ∧ AttendSchool(x))\nInThisClub(bonnie) ∧
     ¬((Attend(x, schoolEvent) ∧
    VeryEngagedWith(bonnie, schoolEvent)) ⊕ (
    Studen(bonne) ∧ AttendSchool(bonnie)))",
 "is_original_correct": "no"}
}
```


Conclusion record (example).


```
{
  "id": "concl_380_1",
  "NL_sentence": "Bonnie performs in school
    talent shows often.",
  "FOL_sentence": "PerformOftenIn(bonnie,
    schoolTalentShow)",
  "label": "Uncertain",
  "corrected": "yes",
  "correction_explanation": "1. Predicate Match:
     Corrected 'Perform' to 'PerformOftenIn' to
    match the premises. 2. Removed Extraneous
    Conjunction: The original included '
    InThisClub(bonnie)' which is not stated in
    the conclusion NL.",
  "FOL_sentence_old": "InThisClub(bonnie) ∧ (
    Perform(bonnie, schoolTalentShow))",
  "label_old": "Uncertain"
}
```


F Example of Released MALLS_TEST_CURATED Dataset Example


```
{
  "NL_sentence": "A vacation is relaxing if it
    includes beautiful scenery and enjoyable
    activities.",
  "FOL_sentence_old": "∀x (Vacation(x) ∧
    Relaxing(x) → (BeautifulScenery(x) ∧
    EnjoyableActivities(x)))",
  "FOL_sentence_new": "∀x (Vacation(x) → ((
    BeautifulScenery(x) ∧ EnjoyableActivities(x)
    ) → Relaxing(x)))",
  "corrected": "yes",
  "correction_explanation": "1. The original
    formula inverted the implication symbols.",
  "id": 1
}
```


G Classification of Common Ambiguities

We report a classification of the recurring NL ambiguities we detected during human oversight of

FOLIO and MALLS. Each type is described with an explanation, a concrete example showing the two plausible FOL readings, and the canonicalization convention we adopt. This is not intended as a comprehensive taxonomy of NL ambiguity in general; it is an empirically grounded catalogue of the patterns that arise frequently in existing NL–FOL datasets for the autoformalization task.

For consistency, in the dataset curation procedure we adopt one interpretation for each category of ambiguity. Nevertheless, since these instances are flagged as ambiguous, users who wish to work with the dataset may choose to resolve ambiguous sentences differently from the choices we adopted here.

`some_plural`. The word *some* followed by a plural noun is ambiguous between asserting the existence of at least one individual satisfying a property and asserting the existence of at least two distinct individuals. For example, *some pets are not mammals* can be formalized as $\exists x\,(\text{Pet}(x) \wedge \neg\text{Mammal}(x))$ or as $\exists x\,\exists y\,(\text{Pet}(x) \wedge \neg\text{Mammal}(x) \wedge \text{Pet}(y) \wedge \neg\text{Mammal}(y) \wedge x \neq y)$. We always adopt the one-witness reading (first formula) and flag the instance as ambiguous. We decide to flag this form of ambiguity since the original FOLIO dataset sometimes use the one-witness formulation and others the two-witnesses one.

`either_or`. The connective *either…or* is ambiguous between inclusive disjunction ($\vee$) and exclusive disjunction ($\oplus$). For example, *Peter's pets are all either monkeys or birds* can be read inclusively as $\forall x\,(\text{PetOfPeter}(x) \rightarrow (\text{Monkey}(x) \vee \text{Bird}(x)))$ or exclusively as $\forall x\,(\text{PetOfPeter}(x) \rightarrow (\text{Monkey}(x) \oplus \text{Bird}(x)))$, where the latter rules out any pet being both a monkey and a bird. We always adopt the exclusive reading ($\oplus$) and flag the instance as ambiguous.

`option_partition`. Enumerated categorization (e.g., *X can be categorized as A, B, and C*) is ambiguous between an inclusive partition (allowing overlap) and an exclusive partition (enforcing disjointness). For example, *machine learning algorithms can be categorized as supervised learning, unsupervised learning, and reinforcement learning* can be read inclusively as $\forall x\,(\text{MLA}(x) \leftrightarrow (\text{SL}(x) \vee \text{UL}(x) \vee \text{RL}(x)))$, or exclusively by additionally enforcing pairwise disjointness. We flag the instance as ambiguous and adopt the inclusive reading ($\vee$), except when it is explicitly stated that the options are incompatible.

`sufficient_necessary_condition`. Conditional phrasing such as *X if P* can denote a sufficient condition ($P \rightarrow X$) or be intended as a biconditional ($X \leftrightarrow P$) when the intended meaning of the NL sentence is to *define* a specific category of entities. For example, *An organization is non-profit if it has a charitable mission, does not distribute profits to owners, and primarily relies on donations* can be interpreted as $\forall x\,((\text{Organization}(x) \wedge \text{CharitableMission}(x) \wedge \neg\text{DistributesProfits}(x) \wedge \text{ReliesOnDonations}(x)) \rightarrow \text{NonProfit}(x))$, but a plausible alternative is a biconditional that also constrains *all* non-profits to satisfy those properties. We flag such cases as ambiguous and, when not clearly disambiguated by the context, adopt the $P \rightarrow X$ interpretation.

`determinative_the`. The definite article *the* presupposes a unique, contextually salient referent, but in isolation it is often unclear whether the formalization should encode uniqueness explicitly. For example, *the cube is red* could be formalized as a simple existential $\exists x\,(\text{Cube}(x) \wedge \text{Red}(x))$, or with an explicit uniqueness constraint $\exists x\,(\text{Cube}(x) \wedge \text{Red}(x) \wedge \forall y\,(\text{Cube}(y) \rightarrow y = x))$, or even as a direct predication over a named constant when the referent has been introduced earlier in a story. Without prior discourse context, the intended reading is underspecified. We flag such cases as ambiguous and, when a single formalization is required, default to the simple existential unless story-level context motivates the uniqueness reading.

`existential_universal`. Bare singular noun phrases of the form *a C has property P* are ambiguous between a universal reading (every C has P) and an existential one (some particular C has P). For example, *a child has a mother* is most naturally read as a universal generalization $\forall x\,(\text{Child}(x) \rightarrow \exists y\,(\text{Mother}(y, x)))$, yet without context, the same syntactic pattern could refer to a specific child. We flag such cases as ambiguous and adopt the universal reading when the sentence expresses a general rule, falling back to the existential reading when the sentence clearly introduces a particular individual.

H Experimental Setup

This section specifies the experimental configuration for evaluating both pipelines. We bench-

mark three LLMs using four prompting strategies (B1–B4) and four prompt variants (pv1–pv4). All combinations are evaluated on FOLIO_VALIDATION, MALLS_TEST, and GGC. We describe the prompting strategies (Section H.1), the prompt variants (Section H.2), the models (Section H.3), the data splits (Section H.4), and how the logical equivalence check is performed (Section H.5). All experimental artifacts are released in the accompanying code repository.

H.1 Prompting Strategies (B1–B4)

We benchmark LLMs using four prompting strategies. As defined in Section 3.1, the model receives p, φ, and Ω, and must output a verdict v alongside a refinement ψ.

B1: Zero-Shot Chain-of-Thought. (Wei et al., 2023). Task description and CoT instruction only; no in-context examples. The model reasons step-by-step over (p, φ, Ω) before producing a verdict and refinement. Establishes a floor for instruction-following and self-directed reasoning alone.

B2: Few-Shot (no CoT). (Brown et al., 2020). Task description plus 5 labeled examples covering all verdict classes, each with its ontology. No reasoning chain is requested. Tests whether exposure to demonstrations alone suffices.

B3: Few-Shot + CoT. Combines B2's in-context examples with B1's CoT instruction.

B4: Feature-Based Decomposition. A form of metaprompting (Zhang et al., 2024). Rather than a holistic judgement, the model evaluates φ against a checklist of four subproperties—*syntactic well-formedness*, *ontological faithfulness*, *truth-conditional equivalence*, and *ambiguity resolution*—before aggregating. Tests whether decomposition improves precision on subtle cases.

H.2 Prompt Variants (pv1–pv4)

For each prompting strategy B1–B4, we render four prompt variants that vary along two dimensions: the evaluator's stance toward the candidate formula and the specificity of evaluation guidance.

pv1 – *neutral, minimal.* A direct task definition positioning the model as an expert linguist-logician. No explicit guidance on skepticism or evaluation protocol; the model is expected to assess the formula on its own merits.

pv2 – *cautious, specified.* Introduces explicit skepticism, instructing the model to "avoid false confidence" and to default to '?' when uncertain. Any doubt about expressiveness, interpretation, or logical equivalence should trigger the uncertain verdict rather than a binary yes/no judgment.

pv3 – *neutral, detailed.* Similar to pv1 in stance, but instructs the model to produce a thorough analysis of the strengths and weaknesses of the candidate formalization before committing to a verdict, encouraging deeper reasoning prior to the final judgment.

pv4 – *adversarial, rigorous.* Adopts the most skeptical stance: the model is instructed to assume φ is incorrect until an adversarial analysis fails to find any flaw. The model systematically probes for common error types (wrong quantifier, scope inversion, predicate mismatch) and assigns yes only after such a search yields no flaw.

All four variants maintain identical task definitions and examples (where applicable) across the three models, ensuring that cross-model comparisons isolate model capability from prompt engineering. We show the first prompt version (pv1) for the B1 prompting strategy below. The full prompt texts are released with the code: the different blocks that builds the prompt combinations discussed above are available at the script `/core/strategies/prompts/common.py`; the examples used in the few-shot version at `/core/strategies/prompts/few_shot_examples.py`.

Prompt template (B1-pv1)

You are an expert linguist and logician evaluating whether a candidate First-Order Logic (FOL) formula $\hat{\varphi}$ correctly and completely formalizes a natural language sentence p using only the symbols of a given ontology.

Assign a quality verdict $q \in \{$'yes', '?', 'no'$\}$:

yes $\hat{\varphi}$ correctly and completely captures the semantic meaning of p within the provided ontology as a valid FOL formula. Note that the chosen formalism and ontology restrictions limit expressiveness: there will be cases in which not all the information in the NL sentence can be retained when converting it into a FOL formula using the given symbols. In such cases, the formalization should nonetheless be considered correct if $\hat{\varphi}$ is the best possible approximation of p in FOL given the provided ontology.

Strategy	Description
B1: Zero-Shot CoT	Task description and step-by-step reasoning instruction only (Wei et al., 2023).
B2: Few-Shot	Task description plus labeled examples covering all verdict classes (Brown et al., 2020).
B3: Few-Shot + CoT	Combines B2's in-context demonstrations with B1's reasoning request.
B4: Decomposition	Metaprompting checklist (Zhang et al., 2024) focused on syntax, ontology coherence, equivalence, and ambiguity.

Table 3: Overview of the four benchmarked prompting strategies.

no $\hat{\varphi}$ is incorrect or incomplete due to syntax errors, semantic hallucination, ontology mismatch, or logical flaws. The only symbols permitted are the logical ones ($\forall\ \exists\ \neg\ \wedge\ \vee\ \rightarrow\ \leftrightarrow\ \oplus\ =\ \neq\ \top\ \bot$), and those explicitly defined in the provided ontology.

? A confident and definitive verdict cannot be assigned because the NL sentence admits multiple non-equivalent interpretations, or because there is uncertainty about its logical correctness, or for other reasons. In this case, your comment MUST begin with exactly one of the following tags: [AMBIGUITY], [UNCERTAINTY], or [OTHER].

When q is "no" or "?", also produce a corrected formalization ψ that addresses the identified issues. ψ should be a rigorous, improved version of $\hat{\varphi}$ that resolves the specific problems you found. Ensure ψ is a complete well-formed FOL formula, not a partial suggestion; when p is ambiguous, ψ should represent the other possible interpretation of p. If q is "yes", put "null" for ψ.

Before producing the verdict, reason step by step.

H.3 Models

We benchmark three LLMs spanning a closed-weights commercial system, an open-weights mixture-of-experts model, and an open-weights dense model:

GPT-4o-mini OpenAI's small frontier model, accessed through the OpenAI API (version gpt-4o-mini-2024-07-18). Training on API inputs was disabled to prevent the model from being updated with data from the private GGC dataset.

Qwen3-30B-A3B. Alibaba's 30B-parameter mixture-of-experts model with 3B active parameters per token, run locally. Extended chain-of-thought reasoning (thinking mode) was enabled.

Gemma 4 31B-it. Google's 31B-parameter instruction-tuned dense model, run locally. Extended chain-of-thought reasoning (thinking mode) was enabled.

For GPT-4o-mini and Qwen3-30B-A3B, the maximum output token limit was set to 2,500, with retries capped at 8,000 tokens. For Gemma 4, the limit was increased to 8,000 tokens – and retries to 13,000 – to accomodate the longer reasoning traces produced during the thinking phase.

We performed the local experiments on a Dell PowerEdge R750 running Red Hat Enterprise Linux 8.7 (Ootpa). The system is equipped with two Intel(R) Xeon(R) Platinum 8360Y CPUs at 2.40 GHz (72 physical cores in total), 512 GB of RAM, 4 TB of disk storage, and an NVIDIA A100 GPU with 80 GB of VRAM.

H.4 Data Splits

Both pipelines are evaluated on the three curated datasets: FOLIO_VALIDATION (275 instances), MALLS_TEST (100 instances), and the GGC dataset (213 instances).

H.5 Logical Equivalence and SMT solver

All logical equivalence checks between FOL formulas are performed using the Z3 SMT solver (De Moura and Bjørner, 2008), with a 30-second wall-clock timeout per query. This configuration is used uniformly for the re-evaluation of LLM performance on the curated datasets, the pipelines evaluation, and NLI label verification. Queries that exceed the timeout are mapped to an `SolverTimeoutError` and excluded from metric computation.

Z3 is invoked under two configurations depending on the dataset:

Raw (FOLIO and MALLS). No additional axioms; equivalence is decided as a pure first-order question over the formulas themselves.

With axioms (GGC). Since GGC formalizes objects in Tarski's World, the relational symbols used are governed by domain constraints. 41 axioms of the domain (symmetry of converses such as Smaller/Larger, the symmetry and reflexivity of SameSize, SameShape, SameCol, SameRow, the mutual exclu-

sivity of shape and size classes, and statements like $(\mathsf{Larger}(x, y) \vee \mathsf{Larger}(y, x)) \leftrightarrow \neg\mathsf{SameSize}(x, y)$) are added as background theory. This axiom set is released with the code.

Timeout errors arise exclusively on GGC, likely due to the added axioms increasing solver complexity; they affect only 0.4% of GGC instances. Notice that timeout errors have no effect for the results discussed in Section 2.3.

I Justification of the Human-Review Prioritisation Order

I.1 Motivation

Each pipeline does not itself constitute a curation strategy; however, the verdict v carries information that can be exploited to define a principled inspection order, directing human reviewers toward the instances most likely to require correction. The rationale rests on a key empirical claim: that accuracy varies systematically with the verdict produced by the V&R stage. Figure 4 provides the evidence for this claim.

I.2 Accuracy by Verdict Group

yes. Instances assigned verdict *yes* achieve the highest $\hat{\psi}$ accuracy in every model–dataset–pipeline cell. Averaged over all 9 cells, Pipeline 1 reaches 0.933 and Pipeline 2 reaches 0.843. Despite the gap between Pipeline 1 and Pipeline 2, even Pipeline 2's *yes* group is substantially more reliable than the other verdict groups, validating the policy of deferring those instances to the last tier of human review.

no. Averaged over all 9 cells, Pipeline 1 reaches 0.514 and Pipeline 2 reaches 0.45. These low values reflect two failure modes: (i) the judge correctly flags an erroneous formula and proposes a refinement that is itself incorrect, and (ii) the judge incorrectly flags a correct formula as *no* and proposes an unnecessary (and potentially wrong) refinement. Both cases result in $\hat{\psi} \not\equiv \varphi_{\text{gold}}$ and therefore require human review, confirming that *no*-verdict instances should be prioritised.

?. Averaged over all 9 cells, Pipeline 1 reaches 0.283 and Pipeline 2 reaches 0.308. Crucially, ? accuracy is lower than no accuracy in most of the cells, empirically establishing that ? instances should be reviewed before no instances in the queue.

Unparseable outputs. A small number of model outputs cannot be assigned to any verdict category because they do not match the expected structured format. Across all 9 cells the total count of such *unparseable* responses is 243. Pooling all cells, the $\hat{\psi}$ accuracy for unparseable outputs is $131/243 \approx 0.54$.

This figure overstates reliability: when the judge produces an unparseable output, the pipeline falls back to the original formula (φ in Pipeline 1, and $\hat{\varphi}$ in Pipeline 2) as $\hat{\psi}$ (Section 3.2), so this accuracy merely reflects how often the original formula happened to be correct. Unparseable outputs provide no information about whether correction is needed and should be flagged for immediate review regardless of apparent accuracy.

Summary of Prioritisation Order Combining the observations above, the pipeline's review order is:

1. **Unparseable**: outcome unknown; review unconditionally.
2. **?**: lowest accuracy; high probability of error or genuine ambiguity requiring human adjudication.
3. **no:** moderate accuracy; likely erroneous but partially addressed by the proposed refinement.
4. **yes:** highest accuracy; the judge is confident and usually correct; review only if capacity allows.

This ordering directly determines the shape of the human-oversight curve: the steepest accuracy gains per reviewed fraction occur in the first tiers, so the area under the curve (AUC) rewards pipelines that concentrate errors early in the queue.

J Other Prioritization Strategies

In the paper we presented a prioritization strategy based on the verdict of the V&R task; one may wonder whether other, more naive strategies could work just as well or better.

The first idea that comes to mind is to order dataset instances from the most difficult to the easiest: presumably, mistakes concentrate more on difficult instances than on easy ones. In the context of formalization it is not clear what "difficulty" means, but as a proxy ordering one can use the length of the NL sentence or of the FOL formula. To make

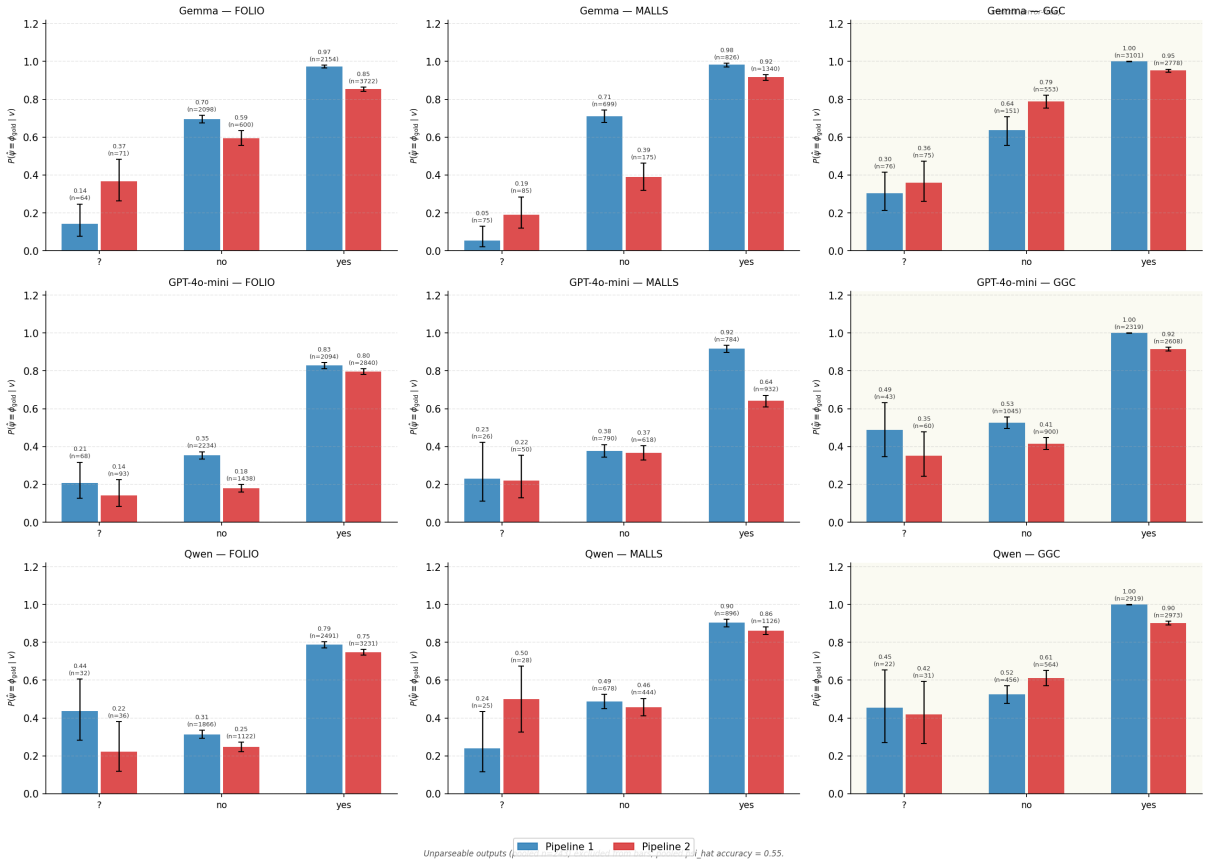


Figure 4: Accuracy conditioned on the verdict assigned by the judge, for Pipeline 1 (blue) and Pipeline 2 (red). Each bar pools all eligible variants (B1–B4 × pv1–pv4 for P1; self-judge B1–B4 for P2) within each cell, yielding large counts and narrow confidence intervals Error bars are 95% Wilson confidence intervals. GGC cells are shaded: because the original dataset is effectively error-free, every Pipeline 1 instance labelled *yes* by the judge corresponds to a correct formula, making the conditioned accuracy equal to 1.00. Unparseable outputs (total pooled $n = 243$) are excluded from the bars because individual cells contain too few observations for reliable estimates.

this metric more robust to predicate/constant renaming, we count the number of *words* for the NL sentence, and the number of *logical elements* for the FOL formula, i.e., quantifiers (together with the variable they bind), logical operators, predicates, variables, and constants.[9]

This length measure induces a prioritization strategy when interpreted as a proxy for instance complexity: a human reviewer could curate the dataset starting from the longest instances and proceeding toward the shortest ones. This strategy can be applied either directly to the original (uncurated) dataset, or after an LLM has already produced a formalization of each NL sentence, in which case the reviewer orders instances by the length of the LLM-generated formula rather than the original one.

Figure 5 compares these two length-based strategies (using both NL length and FOL length) against the V&R prioritization strategy applied with Pipeline 1 and Pipeline 2 (Gemma), for FOLIO and MALLS. Using the convention used throughout the paper, in black we show the case in which we start from the original dataset, and in green when we start from the LLM (Gemma in this case) translations.

The figure shows that, despite these measures seem to correlate with the presence of annotation errors (the slope of the curves is bigger at the beginning and then decreases) our proposed V&R approach outperforms these length-based strategies by a wide margin.

[9]For example, under this notion of length the formula $\forall x(Human(x) \rightarrow Mortal(x))$ has length 6: $\forall x$, $Human$, x, $\rightarrow$, $Mortal$, x.

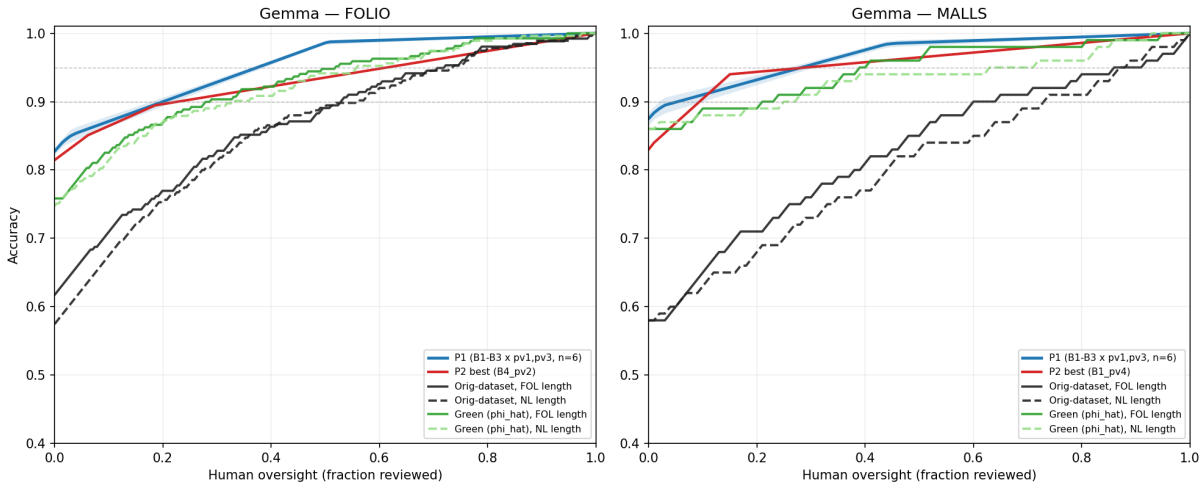


Figure 5: Comparison between Pipeline 1 (blue), Pipeline 2 (red), FOL and NL length based approach starting from the original dataset (black) and from the LLM (Gemma) formalizations (green).

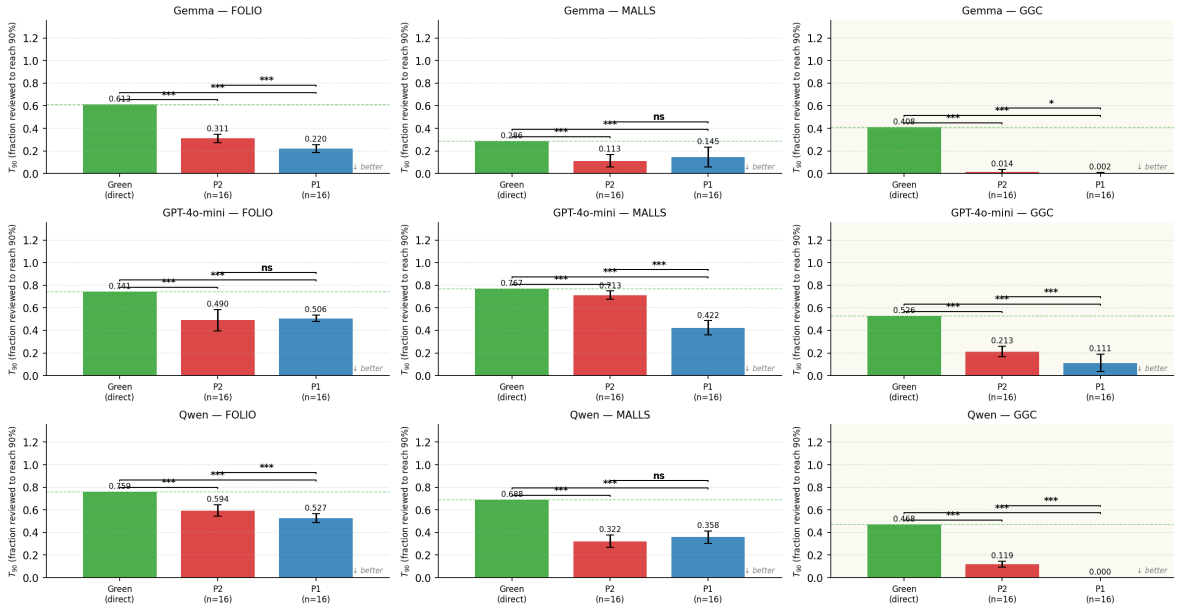


Figure 6: Pipeline comparison across models and datasets under the T_{90} metric.

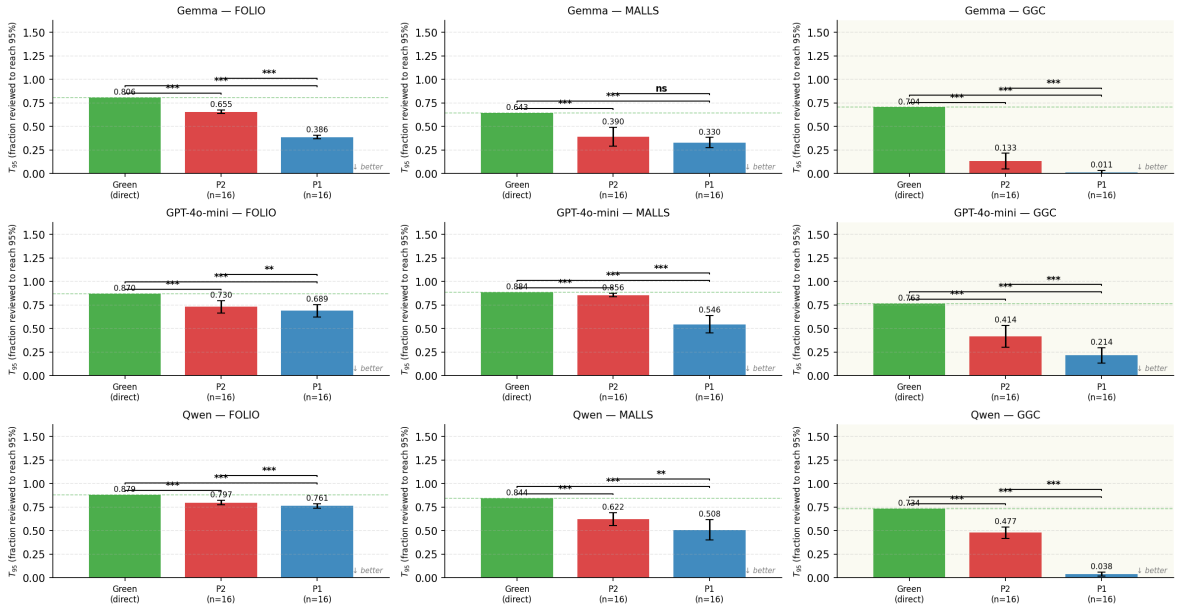


Figure 7: Pipeline comparison across models and datasets under the T_{95} metric.

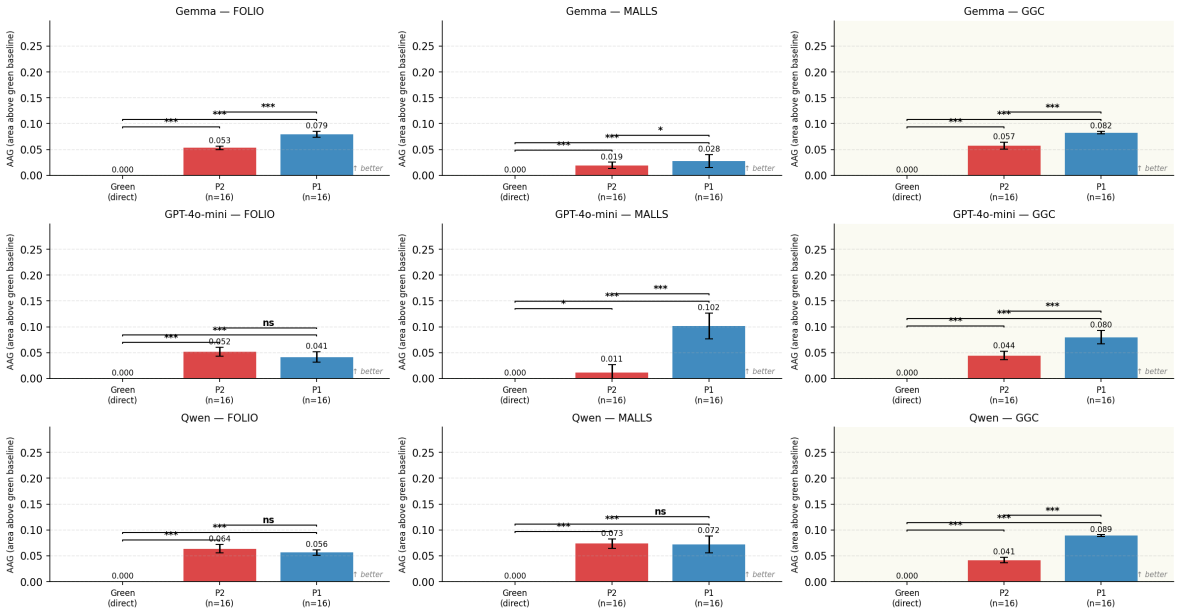


Figure 8: Pipeline comparison across models and datasets under the AAG metric.

K Pipeline Comparison: T_{90}, T_{95}, and AAG

The main paper reports AUC (area under the curve) as the primary scalar summary of pipeline quality. Figure 6-7 replicate the same 3×3 bar grid (model × dataset) for the other complementary metrics T_{90} (fraction of the dataset that must be reviewed to reach 90% accuracy), and T_{95} (fraction to reach 95% accuracy).

Figure 8 complements the analysis with the additional *Area Above Green* (AAG) metric: the area of the region in the accuracy–human effort plane that lies between the Green Baseline curve and the curve of the strategy under evaluation. By construction, the Green Baseline has AAG $= 0$, so every positive value directly quantifies the net gain over starting the curation from the datasets obtained by the direct LLM translation.

All statistical tests use one-sided paired Wilcoxon signed-rank matched on prompting strategies × prompt variants within each cell ($n = 16$ pairs); no multiple-comparison correction is applied within a cell.

The T_{90} and T_{95} bars (Fig. 6–7) confirm the AUC picture from a different angle: Pipeline 1 consistently requires reviewing a smaller fraction of the dataset to reach the accuracy thresholds, and the gains are statistically significant in almost all cells.

L Model Comparison on Pipeline 1

Figure 9 supports the model comparison discussed in the main paper, showing results separately for all three datasets and including the AAG metric (defined in Appendix K). Each bar represents the mean ± std over all 16 prompting combinations (B1–B4 × pv1–pv4) of Pipeline 1.

Gemma's advantage over the other models is consistent across FOLIO and MALLS; on GGC the gap narrows because all models operate near ceiling, as expected for an error-free control dataset.

The AAG results require careful interpretation. On MALLS and GGC, Gemma's AAG is lower than that of the other models, which may seem inconsistent with its overall superiority. The key is that the Green Baseline is model-dependent: it is anchored to each model's own direct-translation accuracy. AAG therefore measures how much a pipeline improves over that model's own baseline, not over a common reference. A stronger model has a higher Green Baseline to beat, so the same absolute accuracy gains translate into a smaller AAG. This is a property of the metric, not a failure of the pipeline.

M Variant Selection and Sensitivity

The main paper restricts the Pipeline 1 analysis to the subset {B1, B2, B3} × {pv1, pv3}, excluding B4 and pv2/pv4. This appendix documents the statistical justification for that restriction and shows that no analogous restriction is warranted for Pipeline 2.

Figures 10 and 11 show 4×4 heatmaps (prompting strategies × prompt variants) of AUC, T_{90}, and T_{95} for Gemma on Pipeline 1 and Pipeline 2, respectively, averaged over FOLIO and MALLS. In Figure 10, the blue-framed cells mark the retained group, which is statistically equivalent within itself and superior to the excluded configurations (B4 and pv2/pv4); higher (lighter) cell values indicate worse performance, confirming that the statistical finding matches the visual pattern.

We use a two-stage non-parametric approach. In stage 1, a Friedman test (non-parametric repeated-measures ANOVA) tests the global null that all levels of a factor are exchangeable. The blocking structure is: for the prompt strategy factor, each block is one (dataset × prompt variant) combination (8 blocks, 4 levels); for the prompt-version factor, each block is one (dataset × prompting strategy) combination (8 blocks, 4 levels). In stage 2, all $\binom{4}{2} = 6$ pairs are tested with two-sided paired Wilcoxon signed-rank; p-values are Bonferroni-corrected by multiplying by 6.

Pipeline 1. The Friedman test is significant for both the prompting strategy factor (AUC: $\chi^2 = 14.55$, $p = 0.002$; T_{90}: $p = 0.001$; T_{95}: $p = 0.002$) and the prompt variant factor (AUC: $\chi^2 = 11.25$, $p = 0.010$; T_{90}: $p < 0.001$; T_{95}: $p = 0.003$). Post-hoc tests identify B4 as significantly inferior to B2 and B3 ($p_{\text{bonf}} \leq 0.047$), and pv2 and pv4 as significantly inferior to pv1 ($p_{\text{bonf}} \leq 0.047$). The pairs B1–B2, B1–B3, B2–B3 and the pairs pv1–pv3 are all non-significant, justifying retaining all three and both prompt variants in the best-combination group.

Pipeline 2. The Friedman test is non-significant for both the prompting strategy factor ($\chi^2 = 1.85$, $p = 0.583$) and the prompt variant factor ($\chi^2 = 0.15$, $p = 0.985$), with all post-hoc pairs also non-

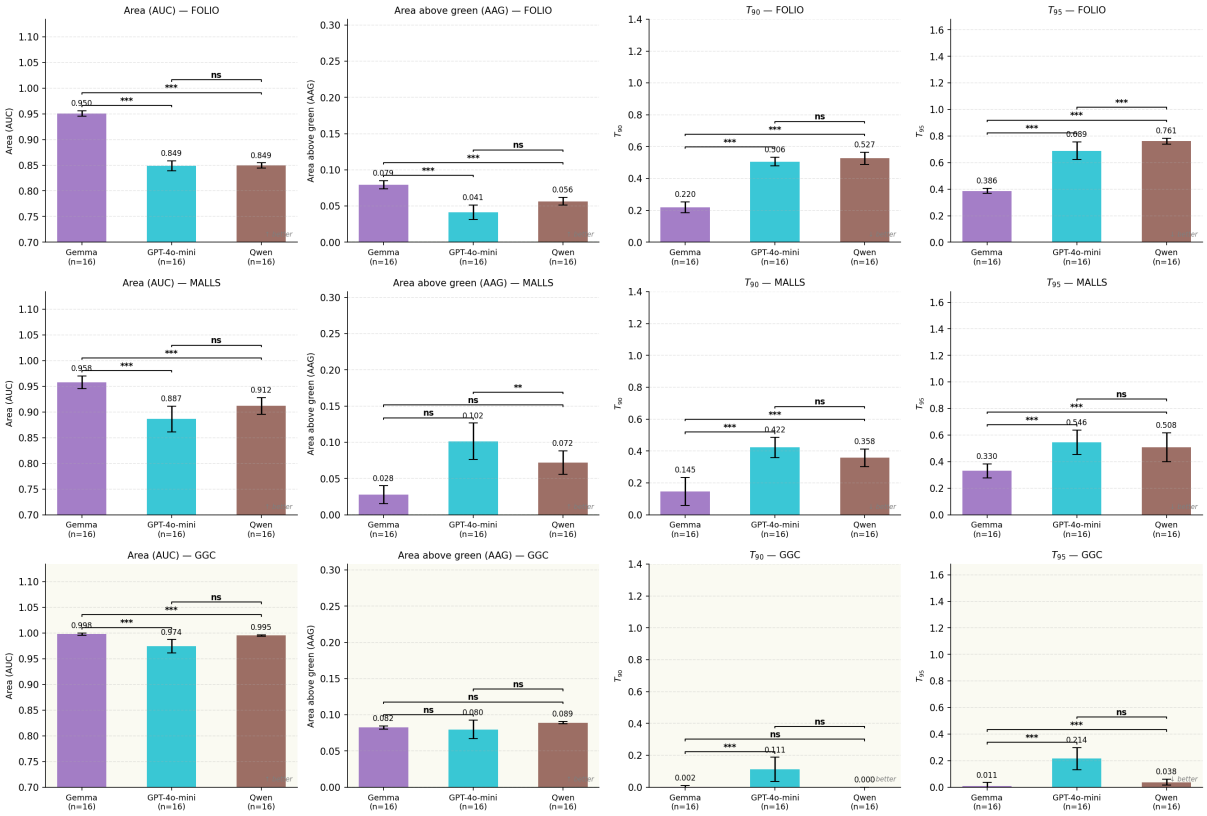


Figure 9: Model comparison across all three datasets and all four metrics (AUC, AAG, T_{90}, T_{95}) for Pipeline 1. Each bar represents the mean $\pm$ std over all 16 prompting combinations (B1–B4 $\times$ pv1–pv4). Brackets indicate pairwise Mann-Whitney U tests with Bonferroni correction (three pairs).

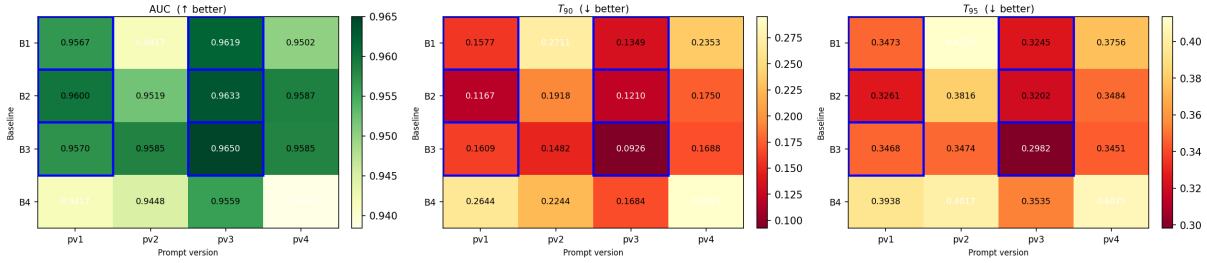


Figure 10: Prompting strategies and variants comparison for Pipeline 1. Blue-framed cells mark the retained group {B1, B2, B3} $\times$ {pv1, pv3}.

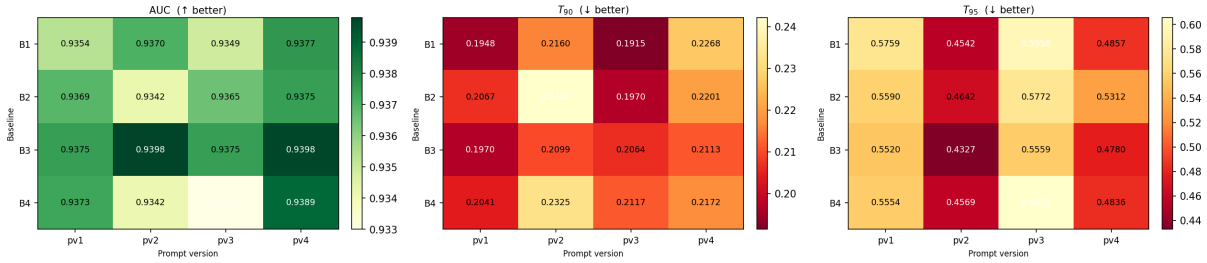


Figure 11: Prompting strategies and variants comparison for Pipeline 2

significant. This indicates that the performance landscape for Pipeline 2 is essentially flat across prompting combinations: no restriction is statistically warranted. For the comparison in the main paper (Figure 3), we therefore do not apply the {B1, B2, B3} $\times$ {pv1, pv3} restriction to Pipeline 2; instead, we select the configuration with the highest AUC, which provides the strongest possible case for Pipeline 2 and makes the superiority of Pipeline 1 all the more notable.

N Oversight Band Grid and Best Variant Results

Figure 12 shows the full 3×3 grid of oversight curves (model×dataset) for Pipeline 1 (blue band), Pipeline 2 (red band), and the direct-translation Green Baseline (green line). The figure generalizes the Gemma-only plot from the main paper (Fig. 3) and supports three observations:

- **Ordering is universal.** The Pipeline 1 $>$ Pipeline 2 $>$ Green Baseline ordering (by AUC) visible for Gemma holds for GPT-4o-mini and Qwen as well on FOLIO and MALLS. The relative gap between Pipeline 1 and Pipeline 2 is largest for Gemma.

- **GGC is near-ceiling for all models.** All three models with Pipeline 1 converge to $\geq$ 0.95 accuracy almost immediately. The oversight gain is real but compressed, confirming GGC's role as an error-free control.

For completeness, Table 4 reports, for each model, dataset, and pipeline, the two prompt configurations with the highest AUC, together with their scores under all metrics considered (AUC, T_{90}, T_{95}). For comparison, we report also for each model and dataset the scores for the Black Baseline and the Green Baseline.

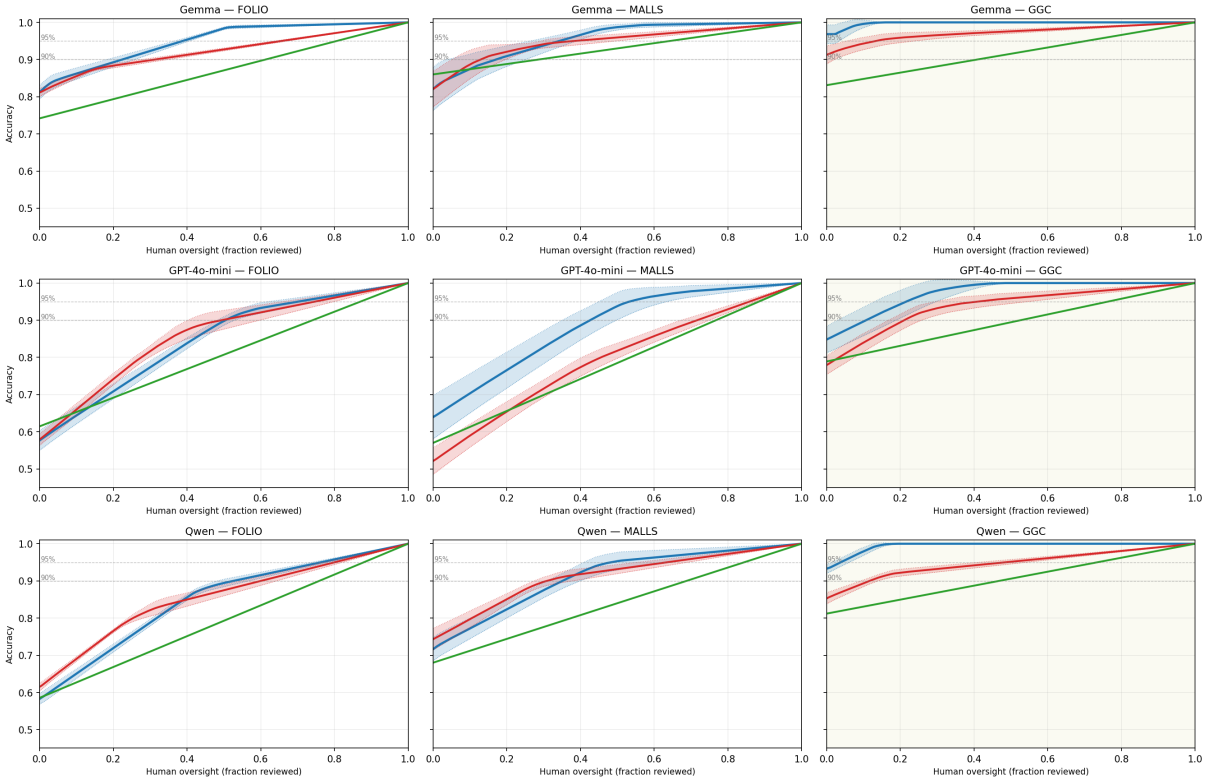


Figure 12: Extension of Figure 3 to all the models and datasets. Each band shows mean $\pm$ std over the 16 variants for Pipeline 1 (blue), and for Pipeline 2 (red); the green line represents the Green Baseline.

Model	Dataset	Pipe	Strategy	Variant	AUC (↑)	T_{90} (↓)	T_{95} (↓)
Gemma	FOLIO	Black	-	-	0.787	0.765	0.882
		Green	-	-	0.871	0.613	0.806
		P_1	B2	pv3	0.959	0.169	0.352
		P_1	B3	pv2	0.958	0.176	0.360
		P_2	B4	pv2	0.931	0.228	0.614
		P_2	B1	pv2	0.929	0.234	0.617
	MALLS	Black	-	-	0.790	0.762	0.881
		Green	-	-	0.930	0.286	0.643
		P_1	B3	pv3	0.974	0.010	0.230
		P_1	B1	pv3	0.969	0.080	0.280
		P_2	B1	pv4	0.957	0.094	0.292
		P_2	B3	pv4	0.957	0.094	0.292
	GGC	Black	-	-	1.000	0.000	0.000
		Green	-	-	0.915	0.408	0.704
		P_1	B2	pv1	0.999	0.000	0.000
		P_1	B3	pv1	0.999	0.000	0.000
		P_2	B2	pv2	0.980	0.000	0.047
		P_2	B3	pv4	0.979	0.000	0.064
GPT-4o-mini	FOLIO	Black	-	-	0.787	0.765	0.882
		Green	-	-	0.807	0.741	0.870
		P_1	B3	pv1	0.873	0.464	0.548
		P_1	B2	pv1	0.861	0.499	0.584
		P_2	B3	pv1	0.873	0.400	0.700
		P_2	B3	pv2	0.869	0.453	0.727
	MALLS	Black	-	-	0.790	0.762	0.881
		Green	-	-	0.785	0.767	0.884
		P_1	B3	pv2	0.910	0.367	0.447
		P_1	B1	pv1	0.910	0.376	0.462
		P_2	B3	pv2	0.830	0.653	0.827
		P_2	B2	pv2	0.825	0.662	0.831
	GGC	Black	-	-	1.000	0.000	0.000
		Green	-	-	0.895	0.526	0.763
		P_1	B4	pv1	0.993	0.000	0.072
		P_1	B4	pv2	0.990	0.000	0.096
		P_2	B2	pv3	0.950	0.170	0.281
		P_2	B1	pv3	0.949	0.173	0.255
Qwen	FOLIO	Black	-	-	0.787	0.765	0.882
		Green	-	-	0.793	0.759	0.879
		P_1	B1	pv3	0.862	0.460	0.719
		P_1	B1	pv1	0.855	0.506	0.753
		P_2	B4	pv4	0.868	0.510	0.755
		P_2	B1	pv4	0.867	0.486	0.743
	MALLS	Black	-	-	0.790	0.762	0.881
		Green	-	-	0.840	0.688	0.844
		P_1	B4	pv4	0.935	0.304	0.413
		P_1	B1	pv3	0.934	0.301	0.395
		P_2	B1	pv3	0.922	0.265	0.562
		P_2	B3	pv3	0.921	0.285	0.529
	GGC	Black	-	-	1.000	0.000	0.000
		Green	-	-	0.906	0.468	0.734
		P_1	B4	pv3	0.997	0.000	0.016
		P_1	B2	pv3	0.997	0.000	0.003
		P_2	B3	pv4	0.955	0.083	0.393
		P_2	B2	pv3	0.954	0.097	0.393

Table 4: Table representing the AUC, T_{90}, and T_{95} of the best 2 variants (according to AUC) for each model, dataset and pipeline